

DISTANCE Chapter 01

The Fake Desktop of Spacetime

Kangbin Yim*

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1 The Most Successful Illusion

We wake up in the morning. The world outside is growing brighter. We check the time, have breakfast, meet people, and begin our work. Before long, the sun descends and evening arrives. We naturally think that another day has passed. The morning has already become the past, the present is now evening, and night will soon follow. When today is over, tomorrow will come. All of this seems perfectly obvious. We naturally arrange the events of a day into a sequence. What happened in the morning came first, what happened at noon followed, and the events of the evening came later. Some things happened only moments ago; others happened years ago. Events that have not yet occurred seem to lie somewhere ahead of us in the future. It is much like a series of gemstones strung along a necklace. If each gemstone represents an event, we tend to imagine those gemstones as being strung one after another along an invisible thread called time. The events are different, but the thread connecting them is one and the same. There are intervals between the gemstones, and we measure those intervals in seconds, minutes, hours, and years. Some events are separated by short intervals, others by long ones. In this way, every event appears to be arranged along a single thread of time, extending from the past through the present and toward the future. This idea is so natural that we rarely even recognize it as an assumption. The same is true of space. There is a cup on the desk in front of us, and a book beside it. A chair stands some distance away. Outside the window there is a building, and beyond it a mountain. We naturally think that all of these occupy different positions within a single space. If we move the cup from

*email: yim@sch.ac.kr (Prof. SCH University, Korea)

the left side of the desk to the right, we think of the same cup as having moved from one position in space to another. The same intuition applies when we leave the room and travel by car. There is a point of departure and a destination. There is a distance between them. The car crosses that distance and moves from one position to another. We think of airplanes, planets, and stars in essentially the same way. There are positions, there are distances between positions, and objects change their positions as time passes. Our everyday world therefore seems to rest on a remarkably simple framework: Objects exist in space, and events occur in time. And when an object changes its position in space over time, we call it motion. What could be more natural than this picture of the world? Human beings probably experienced the world in this way long before mathematics or physics existed. They watched the Sun rise and set and experienced the recurrence of the seasons. They had to judge whether prey was near or far, and how quickly a predator was approaching. They had to estimate how far they needed to travel to find water and whether they could return before sunset. Near and far, fast and slow, before and after were matters of survival long before they became philosophical concepts. Our intuition of time and space is therefore not merely an academic invention. It is deeply embedded in our experience and behavior. From the moment we perceive the world, we are already ordering events, locating objects, and comparing rates of change. Science did not abandon these intuitions. It refined them to an extraordinary degree. Humans who once estimated the time of day from the position of the Sun built clocks. Humans who once estimated the distance between two places by walking developed rulers and instruments of measurement. Over time, they constructed increasingly precise ways of measuring both time and space. Today we can measure intervals far shorter than anything human senses can distinguish. We define time through atomic transitions and electromagnetic processes, and we express astronomical distances numerically. We measure the positions and velocities of objects and use those measurements to predict motions that have not yet occurred. We can calculate where a planet will be at some future moment. We can place a satellite into a desired orbit and send spacecraft toward distant celestial bodies. From processes within atoms to galaxies billions of light-years away, phenomena that appear utterly different can be described within common coordinates of time and space. This success is remarkable. And precisely because it is so remarkable, questioning time and space sounds strange. If concepts allow us to describe and predict so many phenomena with such precision, it seems only natural to regard them as fundamental constituents of reality. Yet the history of science has repeatedly taught us an uncomfortable lesson: What appears most natural

is not necessarily what is most fundamental. The Earth once appeared motionless. This was not merely a foolish illusion held by ignorant people. Even today, when we stand on the ground with nothing but our unaided senses, we barely feel the Earth moving. The ground appears stable, while the Sun emerges in the east and disappears in the west. At night, the stars appear to move across the sky. At the level of direct observation, it is more natural to think that the heavens move while the ground remains still. But when the framework of observation changed, the same phenomena could be explained in an entirely different way. The important point is not that people in the past failed to see the sky correctly. They saw what was actually visible. The problem was identifying the way the world appeared with the way the world was fundamentally organized. Matter provides another example. A stone held in the hand appears to be a single continuous and solid object. Its surface may feel smooth or rough, and we cannot simply push our hand through it. At the scale of everyday observation, there is nothing unreasonable about regarding matter as continuous. But as the scale of observation changed, the appearance of matter changed as well. Atoms were discovered, and within atoms came nuclei and electrons. The atomic nucleus itself did not remain the final unit. As observation reached deeper levels, what we had regarded as a complete object repeatedly appeared instead as the result of smaller constituents and deeper relationships. Once again, the everyday object did not become unreal. A stone is still a stone. We can pick it up, throw it, and injure ourselves if we collide with it. Discovering atoms does not make the stone cease to exist. What disappears is only the assumption that the stone itself must be the deepest possible unit of explanation. Something similar happened in our understanding of life. For a long time, biological species appeared fixed. During the short span of a human life, most organisms retain recognizable forms. Parents resemble their offspring. Horses produce horses, and trees produce seeds of their own kind. But when the scale of observation was extended across much longer periods, species that had appeared fixed could be reinterpreted as temporary forms emerging from long processes of change. Here too, the organisms we directly observe did not disappear. What changed was the scale and perspective through which they were explained. Seen together, these examples reveal an interesting characteristic of scientific progress. Science has advanced not only by discovering things that were previously unknown. It has also advanced by looking again at things we thought we already understood. We abandoned the idea that the Earth occupies the center. We abandoned the idea that species are fixed. We abandoned the idea that matter is continuous. And we abandoned the idea that time and space form an

absolute background identical for every observer. Relativity brought an especially profound change to our intuition of time and space. Time no longer remained an absolute background progressing identically everywhere, and space was no longer a fixed stage shared identically by all observers. Measurements of time and space could vary according to states of motion and physical conditions. And yet one very old assumption has largely remained with us. Time and space may not be absolute, but the universe must nevertheless exist within a fundamental framework of time and space. At this point, we can ask a question. Is the claim that time and space are not absolute the same as the claim that time and space are the most fundamental constituents of the universe? The two claims are not necessarily identical. We may be able to measure something with extraordinary precision. We may even use those measurements to predict phenomena with remarkable accuracy. But success in measurement and prediction does not by itself establish the ontological fundamentality of the concepts used in that measurement. A map provides a useful example. An accurate map can represent real terrain with extraordinary precision. We can measure the distance between two cities on the map and calculate a route to our destination. We can represent elevation, roads, rivers, and mountains. But however accurate the map may be, the coordinates drawn on it do not create the actual terrain. A map is not unrelated to reality. If it were, it would be useless. A good map faithfully preserves important relationships in the terrain it represents. But faithfully representing reality and being reality itself are not the same thing. This distinction matters when we think about time and space. We tend to think that we directly observe time. But what do we actually see? The second hand of a clock moves. A crystal oscillates. The state of an atom changes. The Earth rotates. The seasons change. A person grows older. We compare these changes with one another. We observe how far one process proceeds while another process repeats, and from those comparisons we define seconds, minutes, hours, and other units of time. What, then, have we actually observed? Have we directly observed time itself? We have never seen time emerge from somewhere and cause objects to change. What we actually observe is always change—one state becoming another. Let us return to the necklace. We imagined the gemstones of events as being strung one after another along an invisible thread of time. But what if the only things we can actually observe are the gemstones? There is one event, and then another. One state changes, and another state changes. We compare these changes and assign them an order. Could the independent thread called time, on which we imagine those gemstones to be strung, be not something that exists prior to the events themselves, but rather a scale through

which we order and compare changes? The question may sound strange. We commonly say that “everything changes as time passes.” But what happens if we reverse the presumed causal order in that sentence? Does change occur because time passes, or do we express change through the scale we call time because change occurs? Seen in this way, the role of time changes. Time no longer needs to be an independent flow that causes the universe to change. It may instead be an extraordinarily effective scale for arranging and comparing the progression of different changes. Space can be reconsidered in the same way. Look again at the cup and the book on the desk. We say that the two objects occupy different positions in space. We measure the distance between them. If we move the cup, we say that its spatial position has changed. There is nothing wrong with this description. But again, we can alter the question slightly. Do we actually observe space itself? Can we isolate empty space from everything else and directly observe it as such? What we actually identify are relationships between objects. The cup is to the left of the book. The book is closer to the window than the cup is. When the cup moves, its relationship to the book changes, and so does its relationship to the window. We express these relationships in terms of position and distance. This leaves open another possibility: perhaps space need not be an independent stage that must exist before relationships can exist. Perhaps it can instead be understood as a scale through which relationships among objects are arranged and compared. This alone does not prove that space does not exist. Nor do we need to make such a claim. The purpose of this book is not to eliminate time and space. Time and space are extraordinarily useful. At the scale of everyday life, they are effectively unavoidable realities. We schedule appointments, measure lengths, and calculate distances to destinations. Science and engineering could scarcely function without precise measurements of time and space. There is no reason to deny that practical reality. But we do need to distinguish between what is practically and observationally real and what serves as the most fundamental unit of explanation. A stone can be real without being the most fundamental unit of matter. The Sun can genuinely appear to cross the sky without that observation directly revealing the fundamental structure of motion in the Solar System. Species can be genuinely distinguishable without being eternally fixed entities. There is no obvious reason why time and space should be exempt from the same question. The fact that we observe the universe with extraordinary success through time and space can be distinguished from the claim that, at its deepest level, the universe must be composed of independent entities called time and space. Indeed, perhaps the very success of time and space has made this distinction difficult to notice. If a framework fits

human experience for thousands of years, enables quantitative science for centuries, and continues to operate with extraordinary precision across almost every system of modern engineering, it is only natural that we begin to identify that framework with reality itself. It is in this sense that time and space may be the most successful illusion. Here, illusion does not mean that they are false. It does not mean that we are seeing something that does not exist, like a mirage. In a sense, it means almost the opposite. They work too well. They are too stable. They can be measured too precisely. And for that very reason, we may have come to identify the framework through which we observe reality with the process that actually operates beneath that framework. Perhaps this is better understood not simply as an illusion, but as a remarkably successful form of normalization. It transforms a complex reality into a form that we can understand and measure. By arranging countless forms of change along a common temporal scale, and by organizing countless complex relationships within common spatial coordinates, we become able to compare and calculate the world. Without such normalization, it is difficult to imagine how humans could have developed science as we know it. But the extraordinary success of a normalized representation is not a reason to stop asking what deeper relationships make that representation possible. This is where DISTANCE begins. DISTANCE does not attempt to eliminate time and space. Nor does it deny the achievements of existing physics. It does not claim that observations described through time and space are wrong. Instead, it accepts their success and asks a different kind of question. If time and space are extraordinarily successful scales for describing reality, what changes and relationships actually exist before they are expressed through those scales? Let us reverse the usual order of explanation for a moment. We normally place time in the background first and then say that change occurs within it. But what if change comes first, and time is a scale through which we compare the progression of change? We normally place space in the background first and then say that objects exist within it. But what if relationships come first, and space is a scale through which we observe their arrangement and complexity? Thinking this way does not change the universe we observe. Clocks still work. Planets are still observed along calculable trajectories. Cars still move along roads. GPS still tells us where we are. Starlight still reaches us. What changes is not the phenomenon, but the starting point of explanation. Instead of beginning with time and asking what causes change within it, we begin with change itself. Instead of beginning with space and asking where objects are located within it, we begin with relationships themselves. And then we reverse the question: how do those changes and relationships appear as the observable

world that we describe in terms of time and space? It is too early to answer that question. We have not yet defined Distance, Momentum, or Equalization. Introducing those concepts too quickly would only obscure the significance of the first question by replacing one explanatory framework with another before we have properly examined the one we already use. For now, it is enough to leave open a possibility. Time and space, which we have so naturally regarded as the background of reality, may instead be extraordinarily successful scales through which we observe deeper changes and relationships. A further question then follows naturally. If we cannot directly see the underlying process itself, but only what appears as its observable result, how can we distinguish the world as it is observed from the world as it operates beneath observation? If a representation corresponds to reality so well that its predictions are consistently accurate, can we avoid mistaking that representation for reality itself? And why might human beings have needed such a representation in the first place? To make these questions more concrete, the next section turns to something extremely familiar. A computer desktop. Every day, we move files across a screen, open folders, and delete objects. But the world we see on that screen is not the same as what is actually happening inside the computer. And yet that fake desktop works extraordinarily well. Perhaps that very fact can offer us a useful starting point for thinking again about time and space.

2 The Fake Desktop

When we turn on a computer, a familiar world appears on the screen. There are icons scattered across the desktop. There are folders, documents, photographs, and videos. We select a file with a mouse, move it from one folder to another, and throw unwanted files into the trash. Double-clicking a folder reveals the files inside it; closing the window makes them disappear from view again. All of this feels perfectly natural. There are clearly positions on the screen. One file is on the left, another on the right. Several files appear to be contained inside a folder. When we move a file into another folder, it seems that the place occupied by the file has changed. Copying a file creates another one. Deleting it makes it disappear. We see and manipulate this world every day. Yet inside the computer, nothing happens quite the way it appears on the screen. There are no folders inside the computer that look like the folders we see on the desktop. There are no tiny document-shaped objects sitting somewhere inside a storage device. Moving a file from one folder to another does not mean that a small physical object literally travels from one place inside

the computer to another. What we see is a representation presented to us by the computer. Beneath it, entirely different kinds of change are taking place. States of electronic circuits change. Electrical signals propagate. States in memory and storage devices are altered, and mappings between addresses and data are updated. The operating system modifies information in the file system, while the processor executes countless instructions. Behind the brief visual event of a file entering a folder lie numerous physical and logical state changes that the user neither needs nor is normally able to observe directly. But we do not see any of this. More precisely, we do not need to see it. Instead of displaying the complex internal processes exactly as they occur, the computer translates them into a form that is much easier for us to understand. There are folders. There are files. There are locations. We can grab a file and move it. We can throw away what we no longer need. Through this simple world, we manipulate a much more complex world beneath it. What is interesting is that the desktop's failure to depict the computer literally is not a defect. On the contrary, that is precisely what makes it useful. Imagine that every process actually taking place inside the computer were displayed directly on the screen. Every time we opened a file, changes in memory addresses and page tables would appear before us. Processor instructions would stream continuously across the display. Cache states and bus signals would be shown in real time. We would see which physical locations on a storage device were being read or written, which file-system metadata were being modified, and which system calls were being executed by which processes. Such a display might be much closer to what is actually happening inside the machine. But almost no one would want to use such a computer for everyday work. A representation that is more faithful to the internal process is not necessarily a representation that is more useful to the user. What we want is not to observe everything happening inside the machine at every moment. We want to open a document, edit it, save it, and send it to someone else. So the computer hides its internal complexity. It converts that complexity into a few concepts that humans can easily understand: Object. Position. Distance. Movement. Sequence. A file appears as an object. It occupies a position. A file and a folder appear to have a spatial relationship. A file moves from one location to another. One operation occurs before another. This picture differs considerably from what is actually happening inside the computer. Yet precisely because of this simplification, we can use an extraordinarily complex machine with remarkable ease. This leads to an important distinction. Is the desktop false? In one sense, we could say that it is. The folder on the screen is not an actual container. A file is not a small object shaped like a sheet of paper. The trash is

not a literal wastebasket somewhere inside the computer. Dragging a file to the right does not mean that the data stored inside the machine necessarily move physically to the right. But in another sense, it would be equally misleading to call the desktop simply false. When we delete a file on the screen, something actually changes. When we copy a file, the actual state of the system changes. When we move a folder, rename it, or change its permissions, internal states and relationships within the computer change in correspondence with those actions. The changes on the screen are therefore not empty fiction. They are observable representations corresponding to real internal changes. The desktop is thus both real and not real. More precisely, it is real at the level at which the user interacts with the computer, but it is not identical to the fundamental form of the processes occurring inside the machine. This distinction matters. We do not stop using desktop icons because they are somehow less real than transistors. For the purposes of the user, an icon is in fact a far more immediate reality than a transistor. A person writing a document does not need to know the voltage of a particular memory cell. What matters is knowing which file contains the document. A person looking for a photograph does not need to know which physical blocks of a storage device contain the data. What matters is knowing which folder contains the photograph. To deal successfully with reality, we do not always need direct access to its deepest layer. For a living organism, the ability not to require such access may be even more important. If the Visible World Were All We Could See Consider a more extreme thought experiment. Imagine a person who, from birth and throughout an entire lifetime, can observe nothing but a computer desktop. There is no way for this person to look behind the screen. The processor cannot be seen. Memory cannot be inspected. The storage device cannot be opened. No voltage can be measured, and the internal code of the operating system cannot be examined. The only accessible world is the screen. In that world there are icons and folders. Some icons move. Clicking certain icons opens windows. Moving a file into another folder changes its location. Moving a file to the trash makes it disappear. When storage space runs low, new files can no longer be created. By observing this world long enough, this person could discover extremely sophisticated rules. They could record which actions produce which changes. Experiments could be repeated to test whether the same results recur. The relationship between file size and copying time could be measured. The behavior of different programs could be compared, and the conditions under which particular errors occur could be identified and predicted. With enough accumulated observation and experimentation, this person might construct a highly sophisticated science of the

desktop world. Perhaps equations could even be formulated. The motion of icons under particular conditions might be predicted. The time required to process files of different sizes could be calculated. The behavior of the system as available storage approaches certain limits might be forecast with remarkable accuracy. Would that science be wrong? No. The rules discovered would actually work. The observations would be reproducible, the measurements precise, and the predictions successful. Within the world available for observation, this person could become an excellent scientist. The problem begins only with the next step. Suppose one day the person declares: “The icons, folders, and spatial layout of the desktop are the most fundamental realities of this world.” This is a different kind of claim from the observations that came before it. The fact that icons move according to stable rules is not the same as the claim that icons are the most fundamental entities in the world. No matter how accurate the physics of the desktop becomes, its accuracy does not eliminate the possibility that another layer of operation exists beneath the desktop. Indeed, if the science of the desktop becomes sufficiently successful, the opposite problem may arise. The observer may see no reason to question the screen at all. Everything can be explained. Predictions work. Technology advances. The rules of the desktop can even be used to construct increasingly complex programs. Why, then, suspect that anything exists beneath the screen? At this point, a successful explanation may paradoxically prevent a deeper question from being asked. When an explanation fails, we search for another. But when an explanation works extraordinarily well, we may stop asking what exactly it explains—and what it quietly assumes. This is also why Section 1.1 described time and space as the most successful illusion. Not because they have failed. Because they have succeeded so well. Humans Need a Desktop Too The computer desktop, however, is an interface deliberately designed for the convenience of its users. If the way humans experience the world is also an interface of some kind, who designed it? There is no need to assume a designer. The ways living organisms perceive the world can instead be understood as having taken shape through a history of survival. Did early organisms need an accurate understanding of the deepest structure of the universe? Probably not. They faced more immediate problems. Where is the food? How close is the danger? From which direction is a predator approaching? Should I run now? Which way should I move? With what direction and force should a stone or spear be thrown to reach a target? Can I return to shelter before sunset? Solving these problems does not require direct perception of the deepest physical reality of the universe. What would have mattered far more was the ability to simplify a complex world quickly enough to act.

That is close. That is far. That is approaching. That is moving away. This happened first. That will happen later. If I move now, I will not be too late. Such judgments do not need to reproduce the fundamental structure of the world exactly. They need only be accurate enough for action. A deer escaping a predator does not need an ontology of spacetime. A bird landing on a branch does not need to understand the fundamental nature of space. A human throwing a stone can estimate direction and distance without solving equations of motion in the mind. What a living organism needs is not a complete copy of the world, but a representation of the world that makes action possible. In this sense, the time and space we experience bear an intriguing resemblance to the computer desktop. They organize complex reality into a form in which rapid judgments become possible. They tell us where things are. They distinguish what comes before from what comes after. They allow us to judge what is approaching and what is moving away. And they allow us to anticipate what may happen next if we act in a particular way. That alone provides an enormous survival advantage. There is therefore no obvious reason to assume that human sensory and cognitive systems developed to display the deepest structure of reality exactly as it is. From the perspective of evolution, effective action may have mattered more than an accurate ontology. The world we perceive did not need to be a perfect copy of ultimate reality. It needed to be consistent enough for us to survive within it. Of course, it would go too far to conclude from this that “time and space are fictions created by evolution.” The thought experiment does not justify such a claim. But it does open an important possibility. The form in which the world appears most natural to humans need not be the most fundamental form of the universe itself. Our senses may not simply copy reality. They may normalize it into forms useful for action. In this limited sense, time and space may be regarded as Humanity’s Survival Interface. Interface and Reality

Let us now return to the question raised in Section 1.1. We arrange events in time. We place objects in space. And when an object changes its spatial position over time, we call that change motion. This system works extraordinarily well. What, then, does that success tell us? At the very least, it tells us that time and space correspond deeply to important regularities in the reality we observe. A bad map cannot reliably guide us to a destination. A desktop unrelated to the computer’s internal states could not allow us to control the machine. Time and space therefore cannot simply be dismissed as arbitrary fictions. Yet, as the desktop example shows, corresponding precisely to reality is not the same as being identical to the way reality operates internally. A file moves to the right on the screen. That movement is real. The user genuinely experiences the file as having changed position. And corresponding

changes genuinely occur inside the computer. But none of these facts allows us to conclude that the data physically moved toward the right side of the storage device. “Right” is a direction defined within the interface. “Inside a folder” is a relationship defined within the interface. “Movement” itself is a way in which multiple underlying state changes are represented as one continuous transformation at the user’s level of observation. We can therefore ask at least the same question about physical motion as we experience it. When an object is observed to move from one position to another, must the form of movement we observe be identical to the fundamental form of the change occurring beneath that observation? We do not yet need to answer this question. In fact, it is important not to answer it too quickly. If we push the computer analogy too far, we may drift into an entirely different claim: that the universe itself is a computer. DISTANCE does not require such an assumption. Nor does it require us to suppose that the universe was programmed by someone. Nor does it require the claim that we live inside a simulation. The Fake Desktop thought experiment is intended to establish something much more limited: The consistency and precision of the observable world do not, by themselves, establish that the form of observation is identical to the form of the underlying process. That is enough. Is Time an Interface for Change? We can now apply this principle to time. We express the order of events through time. Morning gives way to noon, spring to summer. Atoms change state, living organisms grow, and stars form and transform. We compare all of these changes along a common temporal scale. This is an extraordinarily powerful normalization. Entirely different kinds of change become comparable in the same units of seconds, minutes, and years. Atomic transitions, human heartbeats, planetary revolutions, and stellar evolution can all be placed on a common scale. Yet, as we asked in Section 1.1, what we directly observe is not time itself, but change. Time may therefore be an interface through which different forms of change become observable in a comparable form. Just as a computer desktop represents many different internal state changes through familiar visual events such as the movement of an icon, time may represent many different kinds of change through a common order and degree of progression. From this perspective, the familiar expression that “time flows” also deserves reconsideration. Is there really some independent entity flowing like a river, carrying every change along with it? Or are there countless changes, whose relative progression we normalize through a common scale and call time? DISTANCE explores the latter possibility. Not in order to eliminate time, but in order to ask what change is before it is expressed through time. Is Space an Interface for Relationships? The same reasoning can be applied to space. We arrange

countless objects within a single spatial framework. Things are near or far. There is left and right. There is above and below. Objects occupy different positions, and distances lie between them. This too is an extraordinarily powerful normalization. Completely different kinds of objects can be placed within the same coordinate system. Their positions can be compared, and their movements can be calculated. But again, what do we directly observe? Relationships between objects. One object affects another. Some relationships are strong, others weak. Some relationships change, while others remain relatively stable. And we express the complex arrangement of these relationships through spatial positions and distances. Space, then, may not necessarily be an empty container given in advance before relationships can exist. It may instead be considered as an interface that normalizes the arrangement and complexity of relationships into an observable form. It is still too early to define Distance here. In particular, the concept of Distance that DISTANCE will later introduce is not identical to geometric distance. Mixing the two at this point would only make the later argument more difficult. For now, one distinction is sufficient. What we observe as spatial distance and the state of the relationship actually operating beneath that observation need not be the same thing. Going Beneath the Screen The Fake Desktop thought experiment leaves us with a methodological warning. No matter how accurately we understand the rules on the screen, that accuracy does not automatically reveal how the system operates beneath the screen. But neither does this make the rules on the screen useless. On the contrary, understanding them with sufficient precision may provide our best starting point for inferring what operates beneath them. Our own situation may be similar. Through time and space, we have measured the universe with extraordinary accuracy. Those measurements are not errors to be discarded. They are among the most important clues we possess. What we can change is the direction of the question. Instead of stopping once phenomena have been described in the coordinates of time and space, we can ask why those phenomena appear in those coordinates in the particular ways they do. To begin asking that question, we must temporarily move the icons of the desktop out of the foreground and look for what is actually changing beneath the screen. At the end of Section 1.1, two things remained: Change and relationships. Something becomes different. And that change has meaning through its relationship with something else. These two remain even when time and space are temporarily removed from the starting point of explanation. The next step, then, is to examine change and relationships more closely. When we compare one change with another, what exactly are we observing? When we say that the relationship between one thing and another has changed, what has actually

become different? And what makes it possible for countless different forms of change and relationship to be represented through the common scales of time and space? As we begin to answer these questions, we can move gradually beneath the familiar desktop of time and space. What we encounter there first is not another kind of object. It is difference—and relationships that change according to that difference.

3 Where Change Begins — Difference

In the previous section, we briefly questioned the surface of the familiar world.

On a computer screen, there are files and folders, each appearing to occupy a certain position. When we drag a file into another folder, it looks as though the file has moved. Yet what actually happens beneath the screen is not the movement of a tiny paper-like object from one place to another. Countless electrical and logical states change, and those changes are represented on the screen in the familiar forms of position and movement.

That does not make the visible world on the screen false.

Files really are opened, copied, and deleted. Changes on the screen correspond to real changes inside the computer. What differs is that the form in which change appears need not be identical to the structure of the change occurring beneath it.

We can ask the same question about time and space.

We arrange events in temporal order and place things within space. When an object changes from one position to another, we call it motion. This framework has worked so accurately and for so long that it is difficult even to imagine the world without time and space.

But as we have already tried, if we temporarily move time and space away from the starting point of explanation, not everything disappears.

Something still changes.

And that change appears through a relationship with something else.

We can therefore move one level deeper.

What must exist for change to be possible?

Can Something Completely Identical Change?

Consider two states, A and B.

Suppose A and B are completely identical in every respect.

There is no way to distinguish them.

If A becomes B, nothing has changed. If B becomes A, the result is the same. From

the states themselves, we cannot even distinguish which came first and which came later.

To speak of change, the before and after must differ in at least some distinguishable way.

If one state becomes another, there must be something that allows the two to be distinguished.

DISTANCE calls this simplest form of non-identity Difference.

Difference is not yet a force or motion.

It is not an object.

It is simply the fact that one state is not identical to another.

Yet this simple fact matters.

Without Difference, there would be nothing by which change could be identified.

If water freezes, there is a Difference between the liquid and solid states.

If the brightness of a star changes, its earlier and later states are not identical.

If a living organism grows, the state before growth is different from the state after it.

If the organization or institutions of a society change, there is a Difference between the relational configurations before and after that change.

Whatever we call change, the concept already assumes that at least two states are not the same.

Difference is therefore not merely an incidental property discovered after change has occurred.

It is a minimal condition that makes change distinguishable as change.

Difference Is Not a Particular Physical Quantity

We should not narrow Difference too quickly into one particular physical concept.

Two objects may have different temperatures.

Two regions may have different pressures.

Their electric potentials may differ.

Their concentrations may differ.

Two objects may have different velocities.

Different materials may exist in different states.

In all of these cases, we can speak of some kind of Difference.

But a temperature difference is not the same thing as a pressure difference.

A potential difference is not the same physical quantity as a concentration difference.

Each is measured in different units and described by different laws.

DISTANCE does not attempt to reduce all of these into one physical quantity.

The word Difference is used here because these different phenomena share a very general condition beneath them.

Two states are not identical.

That is all.

It is important to preserve this simplicity.

If Difference were turned into a new physical substance, we would merely be adding another mysterious entity on top of existing concepts.

There is no Difference particle.

There is no separate force called Difference.

Difference refers to the relational fact that comparable states are not the same.

What that Difference means in an actual physical relationship is the next question.

Where Is Difference?

Consider a red apple.

Is there something called “Difference” stored inside the apple?

That is difficult to say.

The redness of the apple is a state. It becomes a Difference only in relation to something from which it can be distinguished.

Compared with a green apple, there is a Difference in color.

Compared with its own state yesterday, if it has become redder, there is a Difference between temporally distinguishable states.

Compared with its background, there may be a visual Difference.

An isolated state cannot easily be called “different” by itself.

To be different always includes the question: different from what?

Difference is therefore inherently relational.

This point becomes important throughout DISTANCE.

We usually divide the world first into independent objects and then think about relationships among them.

There is a stone.

There is a star.

There is a person.

Only afterward do we consider the relationship between the stone and the ground, between one star and another, or between one person and another.

This order is very natural in everyday experience.

But beginning with Difference opens the possibility of reversing that order.

If Difference itself already requires more than one state, relationships may not be merely secondary features added after fully formed objects already exist.

Even what we identify as a single independent object may contain countless internal relationships.

A stone is not an entirely relationless point.

Neither is a star.

A living organism is even more obviously composed of many relations.

What we identify as one object may be a state in which many components and relationships remain organized in a particular way.

But we do not need to jump from this to the ontological claim that “relationships are more fundamental than objects.”

The point needed here is much more limited.

Difference is difficult to define apart from relationship.

To say that A differs from B already places A and B within some relationship.

Relational Does Not Mean Subjective

There is an easy misunderstanding here.

If Difference requires comparison and relationship, is Difference created by an observer?

Does Difference exist only when someone compares two states?

No.

Two objects at different temperatures can occupy different states whether or not anyone measures them.

A potential difference can exist before a human observer assigns numbers to it.

If non-identical states existed in the early universe, there is no reason to think those Differences had to wait for living observers to appear.

An observer can discover, distinguish, measure, and describe Difference using particular language and numbers.

But the existence of Difference and the description of Difference are not the same thing.

Thus, relational does not mean subjective.

It means that Difference is not a concept completed within one isolated state alone.

At least some relation of distinguishability with another state is required.

An observer may recognize that relation, but need not create it.

This distinction must remain clear throughout Chapter 1.

The possibility that time and space may function as human observational interfaces does not imply that all reality beneath them is created by human perception.

The interface and the change beneath it must remain distinct.

Difference and the Direction of Change

Does the existence of Difference mean that change must immediately occur?

Here we need to be more careful.

The fact that two states differ is not enough to guarantee an immediately observable change.

The world contains too many different things.

A cup on a desk and a distant star are different.

The body temperatures of two people may not be exactly identical.

Two stars may differ in mass, temperature, and composition.

Yet not every Difference in existence immediately changes every other state.

We therefore need to distinguish two statements.

A Difference exists.

And:

That Difference is effective in actual change.

The first states only that two states are not identical.

The second means that the Difference is actually connected to change within a particular relationship.

This distinction is necessary.

Without it, we would be forced toward the strange conclusion that every Difference in the universe acts on every other state with equal significance.

Reality does not appear to work that way.

Some Differences are associated with strong changes.

Some become important only when particular conditions are satisfied.

Others produce almost no observable effect within the current relationship.

Thus, the existence of Difference and the effectiveness of Difference must be distinguished.

That distinction leads directly to the central question of the next section.

Before reaching it, however, we should look a little more closely at how Difference appears in physical change.

Difference and the Energy Gap

In physical change, Difference often appears through differences in energy.

We can consider states at different heights.

We can consider different electrical states.

We can consider two states at different temperatures.

Physics describes each situation precisely using the language of the relevant form of energy and interaction.

Within DISTANCE, the expression Energy Gap becomes important.

But Difference and Energy Gap should not be treated as identical concepts.

Difference is more general.

Difference is the fact that two states are not the same.

An Energy Gap refers to a case in which that Difference is expressed through an energetic relationship.

This distinction becomes especially important when the discussion later extends to life and society.

If every Difference were immediately identified with a physical energy difference, the scope of the framework would become unnecessarily narrow.

In physical phenomena, however, Energy Gap is particularly important.

Energetically non-equivalent states contain an inequality that may be connected to change.

When that inequality becomes effective within an actual relationship, the possibility of change becomes physically significant.

Here too, it is important not to imagine an Energy Gap as a small substance.

There is not some “gap” stored on one side that then travels to the other.

An Energy Gap expresses the fact that related states are not energetically equivalent.

Once again, the starting point is relational.

This later becomes an important foundation for the DISTANCE interpretation of physical motion.

But we do not yet explain here how Energy Gap appears as a directionality of motion.

That belongs later.

Is Change a Process of Eliminating Difference?

When two objects at different temperatures are placed in contact, their temperature difference may decrease.

When regions of different concentration are connected, diffusion occurs.

When a pressure difference exists, flow appears and the difference changes.

Looking only at such cases, we may be tempted to say that change is simply a process

that eliminates Difference.

But that is only partly correct.

While one Difference decreases, other relationships may change.

When a hot object exchanges heat with a cold one, the relationships of both objects with their surrounding environment also change.

When a pressure difference changes in one region, the distribution of matter changes, and new pressure differences may emerge elsewhere.

A change in one state simultaneously alters the conditions of the other relationships in which that state participates.

In the actual world, therefore, Differences are not independently erased one by one.

When one Difference changes, the conditions of other Differences change as well.

This point is crucial.

If only the relationship between A and B existed, the situation would be simple.

A change in the Difference between A and B might be the end of the story.

But if A is simultaneously related to C, D, and E, then a change in A also changes all of those relationships.

While the Difference between A and B decreases, the Difference between A and C may increase.

A previously insignificant Difference may become important.

The transformation of the world as a whole should therefore not be reduced to a simple process in which "Difference continually disappears."

It is closer to a process in which the states and arrangements of Differences continuously change throughout a relational network.

We do not yet need to name the overall direction of that process.

For now, what matters is that the transformation of Difference is not an isolated event.

One Difference and Countless Differences

Let us now move beyond the simplest picture containing only two states.

There is a Difference between A and B.

That gives us one relationship.

But in the actual world, A is not related only to B.

There may be an A-C Difference.

An A-D Difference.

There may also be Differences between B and C, B and D, and C and D.

These relationships need not all be of the same kind.

One object participates simultaneously in many kinds of relationship.

As a result, when one state changes, not one but several relational Differences may change at once.

In this sense, the world resembles a network of relationships more than a simple chain.

Linear descriptions such as “A changed B, and B then changed C” can be very useful for explaining particular situations.

But at a broader scale, A, B, and C may all be changing simultaneously through multiple relationships.

And each change modifies the conditions of still other Differences.

A state at any given moment may therefore be difficult to explain fully through only one preceding cause.

Countless Differences coexist.

The universe that DISTANCE will later consider is precisely such a world of multiple relations.

What we observe as a single star, planet, or living organism does not consist of only one isolated relationship.

There are countless Differences within it, and it also participates in many external relationships.

The simple A-B model used to explain one Difference is therefore only a minimal conceptual unit.

In the actual world, such relationships overlap without number.

Why Difference Matters

We can now return to the question with which this chapter began.

If time and space are temporarily moved away from the starting point of explanation, what remains?

Change and relationship remained.

And when we examined change and relationship more closely, we found Difference beneath them.

For change to exist, states cannot be completely identical.

For a relationship to be distinguishable, there must be something distinguishable among the states that form it.

In physical change, such Difference may appear as an Energy Gap.

But Difference is a broader concept than Energy Gap.

Difference is not a new substance hidden inside an independent object.

It is the fact that states within a relationship are not identical.

Seen this way, Difference is a remarkably modest concept.

And precisely because it is modest, it can serve as a useful starting point for explanation.

We have not yet asked why a particular Difference exists.

Nor have we asked why the universe was not completely identical from the beginning.

Those are not the questions required here.

The claim needed at this point is much more limited.

If observable change exists, then that change contains distinguishable Difference among states.

And those Differences do not exist one by one in isolation.

They coexist within different relationships, and one change alters the conditions of others.

That alone produces the next problem.

Not Every Difference Matters Equally

A differs from B.

C differs from D.

But this does not mean that the Difference between A and B and the Difference between C and D have the same significance in actual change.

Some Differences are connected to immediate change.

Others persist for long periods.

Some strongly affect other relationships.

Others produce almost no effect within the present relational network.

Difference alone is therefore insufficient.

Difference can tell us what is not the same.

But it cannot by itself tell us:

how effective that Difference is within the present relationship.

These are different questions.

And it is precisely at this distinction that the next concept of DISTANCE becomes necessary.

We have now established the existence of Difference.

The next task is to express the degree of effectiveness that a Difference possesses within a relationship.

DISTANCE addresses this through the concept of Distance.

But we do not define Distance here.

For the present section, one conclusion is enough:

Even when Difference opens the possibility of change, not every Difference is equally effective within every relationship.

Why, then, do some relationships act more strongly than others?

How can that difference be expressed?

And what is the connection, if any, between that relational effectiveness and the spatial nearness or farness we ordinarily call distance?

That is where the next section begins.

4 The Distance of Relationships — Distance

In the previous section, we examined Difference as the simplest condition for change.

If two states are completely identical, there is nothing by which a change between them can be distinguished. For observable change to exist, there must at least be some difference between distinguishable states. In physical relationships, such Difference may also appear in the form of an Energy Gap.

But one problem remained at the end of Section 1.3.

The mere existence of Difference is not enough to explain the changes we actually observe.

There are countless Differences in the world.

There are different temperatures, different pressures, different electrical potentials, different concentrations, and different states of motion. More broadly, materials differ in composition, stars differ in condition, and living organisms and environments differ from one another in countless ways.

Yet not all of these Differences are equally effective in every relationship.

Some are connected to strong change.

Some persist for long periods.

Some produce almost no observable effect within a given relationship.

Once Difference has been identified, we therefore need another question:

How effective is that Difference within the relationship?

DISTANCE introduces the concept of Distance to address this question.

The Distance We Already Know

The word distance is, of course, nothing new.

We experience distance every day.

The phone in our hand is close.

The building outside the window is farther away.

Another city is much farther.

Another planet or star may be unimaginably distant.

We measure distance with rulers, maps, radar, and observations of light. Given the positions of two points, we can calculate the geometric distance between them.

This concept is so natural and so successful that we normally begin with space, place two objects within it, and then measure the distance between them.

The order of explanation is usually something like this:

space $\rightarrow$ position $\rightarrow$ distance between objects $\rightarrow$ interaction

When two objects are close, some interactions may become stronger. When they move farther apart, those interactions may weaken.

This framework has been extraordinarily successful in describing the physical world.

DISTANCE does not reject that success.

But, as we have done throughout Chapter 1, we can reverse the order of explanation.

What if space is not placed first?

What if we begin instead with Difference and relationship, and then ask how effective that relationship is in actual change?

What, in that case, would “distance” mean?

This is where Distance in DISTANCE begins.

What If We Think of Relationship Before Spatial Separation?

Suppose there are two states, A and B.

There is a Difference between them.

But as we saw in the previous section, the existence of Difference alone does not tell us whether strong change will occur between A and B.

The Difference may be almost ineffective within their actual relationship.

Or it may be strongly connected to changes in both states.

We therefore need to distinguish at least two questions.

First:

How different are A and B?

Second:

How effective is that Difference within the relationship between A and B?

The first question concerns Difference.

The second is where Distance becomes necessary.

In DISTANCE, Distance does not begin as the length of empty space separating two states.

It is a concept for expressing the effective nearness or farness of a Difference within an actual relationship.

If the relationship is strongly effective, the states are relationally close.

If the relationship is scarcely effective, they are relationally distant.

In its simplest form:

Distance expresses the effective nearness or farness of a relationship.

Here, near and far do not directly refer to geometric position.

They describe the state of a relationship.

We Already Use Relational Distance

This use of the word may initially sound unfamiliar.

Yet in everyday language, we already use distance and closeness in ways far broader than spatial separation.

We say that two people are very close.

We say that two ideas are close to one another.

We describe one event as closely related to another.

We may even say that two people living in the same house have grown distant from each other.

In such cases, nearness and distance are not measured in meters or kilometers.

Two people can occupy the same room and still have a distant relationship.

Two people separated by thousands of kilometers may remain strongly connected and deeply influential to one another.

We therefore already distinguish, intuitively, between at least two kinds of distance.

One is geometric separation.

The other is relational closeness.

DISTANCE does not simply import this everyday metaphor into physics.

But the intuition reveals an important question.

When we describe the relationship between two states, must geometric distance be the only meaningful form of distance?

Or might geometric distance itself be related, in some deeper way, to a more fundamental form of relational nearness?

DISTANCE leaves open the second possibility.

The Same Difference, Different Relationships

We can make the need for Distance more concrete.

Suppose two different situations contain the same temperature difference.

The temperature difference between A and B is twenty degrees.

The temperature difference between C and D is also twenty degrees.

In terms of Difference alone, the two situations appear the same.

Now suppose, however, that A and B can exchange heat very easily, while thermal transfer between C and D is severely restricted.

The temperature Difference is the same in both cases.

But the significance of that Difference for actual change is not the same.

In the A-B relationship, the Difference may be strongly effective.

In the C-D relationship, the same Difference may remain while producing only a limited effect.

The magnitude of Difference alone therefore cannot distinguish the two situations.

We need to distinguish between:

the magnitude of Difference

and

the effectiveness of that Difference within a relationship.

Distance is introduced in DISTANCE to express the second.

This distinction is fundamental.

If Difference and Distance are collapsed into a single concept, it becomes difficult to explain why apparently similar Differences can have very different significance in different relationships.

Near Relationships and Distant Relationships

What, then, does it mean for two states to be relationally close?

At this stage, we do not need a mathematical definition.

A conceptual one is enough.

If changes in A are strongly significant for B, and changes in B are likewise strongly significant for A, then their relationship can be described as close.

If A can change substantially while producing almost no effective difference for B, and B can likewise change with almost no effect on A, then their relationship can be described as distant.

Nearness in DISTANCE is therefore not merely positional proximity.

It is the effectiveness of mutual relationship.

This leads to an important consequence.

The same two objects may have different Distances depending on which relationship is being considered.

They may be strongly connected through one physical relation while remaining almost independent through another.

This becomes even more obvious when we later consider complex living or social systems.

Two people may be economically connected while remaining emotionally distant.

Two countries may be geographically distant while being closely linked through information or economic exchange.

These social examples should not be treated as identical to physical relationships.

But they help illustrate why Distance need not always be reduced to a single geometric separation.

Is Distance a Single Number?

A natural question follows.

Can the Distance between two states always be expressed as one number?

At this stage, there is no need to assume so.

In geometry, it is natural to represent the separation between two points with a single value.

Relational Distance may be more complex.

A and B may participate simultaneously in thermal, electrical, mechanical, or other kinds of relationship.

The effectiveness of Difference may vary across those relationships.

It would therefore be premature to assume that Distance must always be reducible to one universal scalar quantity.

In this book, Distance is initially used as a conceptual measure of relational effectiveness.

Whether a particular physical relationship can later be quantified in a specific math-

ematical form is a separate question.

The purpose here is not to invent a new unit.

It is to reverse the order of explanation.

Instead of assuming geometric space first and then asking how the strength of relationships is determined within it, we begin with relationships and ask how their effectiveness may be expressed in terms of nearness and farness.

But Why Do Things Interact More Strongly When They Are Geometrically Close?

At this point, a strong objection appears.

In the physical world, geometric distance clearly matters.

Many interactions become stronger when objects are spatially closer.

In many cases, the effects of interaction weaken as objects move farther apart.

If so, why distinguish geometric distance from relational Distance at all?

Why not simply say that spatial distance determines the strength of the relationship?

This objection is important.

It is also why DISTANCE does not deny geometric distance.

If geometric distance had no relationship whatsoever to physical interaction, it could not function as such a remarkably successful predictive quantity.

We use distance to calculate planetary motion.

We use it to describe electromagnetic interaction.

We infer relationships among celestial bodies through spatial observation.

We design countless engineering systems in terms of position and geometric separation.

Geometric distance is therefore clearly an important observational scale of physical relationships.

The question posed by DISTANCE is not whether geometric distance is real or useful.

The question is why it is so useful.

That is, the question changes from:

“Is spatial distance real?”

to:

“Why does relational effectiveness correspond so reliably to spatial distance?”

These are different questions.

The first debates the existence of geometric distance.

The second accepts its success and asks whether it is the deepest explanatory level.

A Map Is Not the Terrain, but It Is Not Wrong

We can connect this problem to the interface argument of Section 1.2.

On a map, two cities have a measurable distance between them.

That distance does not capture every detail of the actual terrain.

Elevation, road conditions, traffic, climate, and difficulty of travel cannot all be represented by one straight-line distance.

Yet the distance shown on the map is not therefore wrong.

It represents certain relationships in reality in an extremely useful form.

The computer desktop offered a similar example.

Two icons may appear far apart on the screen, while the corresponding data need not be physically separated in the same manner within memory or storage.

Yet the positions of the icons have precise meaning within the interface.

We can use them to select, move, and organize files.

A representation can differ from its underlying mechanism without becoming false.

Spatial distance may be understood in a similar way.

Even if geometric distance is not identical to relational Distance, spatial measurement can remain entirely valid if there is a stable correspondence between them.

This does not weaken the concept of space.

It asks why spatial measurement works so well.

Can Things Be Spatially Close but Relationally Distant?

We can push the distinction with an extreme question.

Can two things be geometrically very close and yet relationally distant?

Conceptually, yes.

If almost no effective relation exists between two nearby states in some particular respect, the Distance of that relationship may be long.

Conversely, even when geometric separation is large, a particular relationship may remain strongly effective and therefore relatively close in the relational sense.

Of course, in actual physical systems geometric separation is often deeply connected to relational effectiveness.

The two must not be separated arbitrarily.

DISTANCE does not claim:

geometric distance is unimportant.

The more precise claim is:

Geometric distance and relational Distance need not be assumed to be identical concepts from the outset.

Once this distinction is allowed, a new question becomes possible.

Perhaps what we call geometric distance is an especially stable observational expression of relational states.

If so, spatial arrangement may not merely determine relationships; it may also reflect the results of relational processes.

At this stage, opening that possibility is enough.

Energy Gap and Distance

In the previous section, we noted that Difference may appear in physical relationships as an Energy Gap.

How, then, are Energy Gap and Distance related?

Again, we should first keep the concepts distinct.

Energy Gap expresses energetic non-equivalence between states.

Distance expresses how effective that Difference is within the present relationship.

The two are therefore related, but not identical.

The existence of the same Energy Gap does not necessarily imply the same Distance if the relational conditions are different.

Conversely, when a relation is strongly effective, DISTANCE describes that relation as close.

In physical discussion, we therefore need to distinguish among three ideas:

Difference

What is not the same?

Energy Gap

How is that Difference expressed energetically?

Distance

How effective is that Difference within the current relationship?

These concepts are connected, but they are not interchangeable.

This distinction must remain stable if the later argument is to remain consistent.

In particular, the magnitude of an Energy Gap should not simply be renamed Distance.

Distance includes the condition of relationship.

Even a large Difference may have little effectiveness in a particular relation.

A smaller Difference may, under different conditions, participate in a highly effective relationship.

Distance is therefore not merely another name for the amount of Difference.

Nearness Does Not Mean Stability

There is another important misunderstanding to prevent.

In everyday intuition, becoming closer often suggests combination, stabilization, or completion.

Two objects come together.

Two people become closer.

Several things gather into one configuration.

The phrase short Distance may therefore easily be interpreted as meaning a more stable or more resolved state.

That interpretation does not follow in DISTANCE.

Relational nearness means that the relationship is strongly effective.

It does not mean that the relationship has already become stable.

A strongly effective Difference may contain a great deal of potential for change.

Therefore:

short Distance $\neq$ stabilized state

Likewise:

long Distance $\neq$ geometrically far away

Both mistakes would simply impose geometric intuition directly onto relational Distance.

This boundary will become especially important later.

A Note on the Word “Preferred”

A strongly effective relationship may become more significant in actual change.

Later chapters will therefore use expressions such as preferred relationship, preferred path, or preferred Equalization.

The word preferred must be interpreted carefully.

It does not imply that nature possesses intention.

It does not mean that a relationship knows its goal and selects an optimal route.

It does not imply that the universe calculates which path is best.

It means only that when a Difference is more strongly effective within a relationship, that relationship may participate more prominently in actual change.

In this sense, preferred refers to relational priority, not purpose.

We do not explain that process itself here.

The point at this stage is simply to prevent the language of preference from being mistaken for teleology.

How Many Distances Does One Object Have?

Consider a stone on a desk.

We can assign it a position and calculate its geometric distance from surrounding objects.

From the perspective of relational Distance, however, the situation is far more complex.

The stone is related to the Earth.

It is related to the surrounding air.

It is related to light.

It is related to nearby matter.

And the components within the stone are related to one another.

Each of these relationships may contain different Differences.

And those Differences may be effective to different degrees.

It is therefore not sufficient to say that a single object possesses one Distance.

More precisely, an object exists within a network of many relational Distances.

This produces an important shift in perspective.

From a spatial viewpoint, we normally assign an object a position first and then calculate its distances from other objects.

From a relational viewpoint, the network of relationships surrounding the object becomes the prior question.

This may eventually force us to reconsider what it means to identify something as one object in the first place.

But that question comes later.

For now, it is enough to recognize that one object does not participate in only one isolated Distance.

A Network of Distance

A is related to B.

A is also related to C.

B is related to C.

Add D, E, and F, and the number of possible relationships increases rapidly.

Each relationship may contain Difference.

And each Difference may be effective to a different degree.

If reality is described through relational Distance, the world begins to resemble not a simple collection of isolated points, but a network in which many different Distances overlap.

This network need not remain fixed.

When states change, Differences change.

When Differences change, the effectiveness of relationships may change.

Distance, therefore, need not be treated as a permanent property.

At this point, we arrive at an important transition.

If Distance expresses relational effectiveness and relational states can change, then Distance itself can also change.

But the fact that Distance can change still does not tell us the direction of change.

Knowing that one relationship is more effective than another is not the same as knowing what actual change will occur.

That is where the next section begins.

Distance Has No Direction

Consider the geometric distance between two points.

The distance tells us how far apart they are.

But the distance alone does not tell us in which direction either point will move.

Relational Distance is similar.

Suppose the A-B relationship is closer than the A-C relationship.

That tells us that the A-B relationship is more strongly effective.

But it does not yet tell us how A will actually change.

Change requires directionality.

The fact that a Difference is effective within a relationship and the direction in which that relationship tends to change must remain conceptually distinct.

This is the limit of what Distance can explain.

Distance tells us how near or far a relationship is.

It does not yet tell us where that relationship tends to change.

We therefore need another concept.

That concept must not merely restate the strength or nearness of a relationship.

It must address the directionality through which an effective Difference appears in actual change.

DISTANCE calls that directionality Momentum.

We do not define Momentum here.

It is enough to reach the point at which Difference and Distance alone are no longer sufficient to explain the direction of actual change.

In Section 1.3, we asked:

What is different?

In Section 1.4, we asked:

How effective is that Difference within the relationship?

One question now remains.

In what direction does that effective relationship tend to change?

That is the starting point of the next section:

1.5 The Directionality of Change — Momentum.

5 The Directionality of Change — Momentum

In the previous section, we introduced Distance as a concept expressing the effective nearness or farness of a relationship.

The existence of Difference alone was not enough to explain the degree to which actual change may occur. Even identical or similar Differences may have very different effectiveness depending on the relationships in which they exist. We therefore needed Distance to express how strongly a Difference is effective within a relationship.

But Distance alone is still not enough.

Suppose we know that one relationship is closer than another—that it is more strongly effective.

Does that alone tell us what will actually change, or how?

It does not.

Change requires something more than magnitude or effectiveness.

It requires direction.

There may be two states at different temperatures.

There may be two states at different pressures.

There may be states at different heights.

There may be electrically non-equivalent states.

In each case, there is more than the simple fact that the two states are different. If the Difference is effective within an actual relationship, change does not appear randomly in every possible direction. It exhibits a particular directionality.

Difference describes relational non-equivalence.

Distance describes how effective that Difference is within a relationship.

But to describe the direction in which actual change appears, another concept is required.

DISTANCE calls this relational directionality of change Momentum.

A Familiar Word, a Different Level of Question

Momentum is not a new word.

In physics, momentum has long been one of the central concepts with a precise and well-established meaning.

In classical mechanics, it is commonly expressed as

$$p = mv.$$

When an object with mass is moving, its momentum can be represented by the product of its mass and velocity. Conservation of momentum is one of the fundamental principles of physics and provides a powerful way to describe motion, collisions, and interactions.

DISTANCE does not reject this concept.

Nor does it claim that conventional physical momentum is mistaken.

The same word is used here because the concept we are considering is also concerned with the directionality and persistence of change.

The level of explanation, however, is different.

Conventional momentum describes an observable state of motion with great precision. DISTANCE asks a question one step earlier:

Why does change appear in that direction?

Instead of beginning with an object that is already moving, it attempts to descend to the relationships from which observable motion emerges.

Without this distinction, the two meanings of Momentum can easily become confused.

Momentum in DISTANCE is not intended as a new variable that replaces momentum in conventional mechanics. It is a broader relational concept introduced to describe how

directionality emerges from a relationship in which Difference is effective.

Does Something at Rest Have No Momentum?

Consider a stone resting on a table.

Within our observational frame, the stone does not move.

Its velocity is zero.

Its conventional mechanical momentum, $p=mv$, can therefore be represented as zero.

In that specific mechanical sense, it is entirely correct to say:

“The object has no momentum.”

DISTANCE asks a different question.

Have all directionality of change associated with the stone really disappeared?

The stone is related to the Earth.

It is related to the table.

It participates in thermal relationships with the surrounding air.

It interacts with light.

Within the stone itself are countless components and relationships among them.

The fact that the stone does not change its macroscopic position does not mean that all Differences within all these relationships have disappeared.

Multiple relational directions of change may instead constrain or counterbalance one another, or remain organized within a particular configuration, so that no macroscopic displacement appears at our level of observation.

We must therefore distinguish between two statements:

There is no observable change of position.

and

There is no relational directionality of change.

They are not equivalent.

This distinction becomes important throughout DISTANCE.

Rest need not be a special condition in which Momentum is absent.

It may instead be a condition in which multiple Momentum coexist but do not appear, at the present macroscopic scale, as one observable displacement.

Does Directionality Appear Only After Motion Begins?

We often observe motion first and then search for its cause.

A ball falls.

A planet moves.

Water flows.

Objects collide.

We then explain what forces acted, how energy changed, and why the observed motion occurred.

This approach is extraordinarily successful.

From a relational perspective, however, we can reverse the order of the question.

What if the relationship associated with the falling ball already exists before the ball begins to fall?

What if the relevant Difference is already present before water begins to flow?

Consider a drawn bow.

The arrow is not yet flying.

But this does not mean that the bow, string, and arrow contain no possibility of change.

When the string is released, the direction in which the arrow moves is not created from nothing at that instant. It is already related to the configuration of the bow, string, and arrow before release.

Observable motion may therefore be preceded by a relational state that already makes a particular direction of change possible.

DISTANCE does not treat Momentum merely as something produced by motion.

Rather, when directionality already present within a relationship becomes effective, we may observe it as motion.

The familiar explanatory order might be written as:

motion $\rightarrow$ Momentum

DISTANCE instead considers:

Momentum $\rightarrow$ observable change

The arrow here does not represent a simple temporal sequence.

It does not mean that Momentum is first produced as some substance and then pushes an object into motion.

The claim is more limited:

Relational directionality is explanatorily prior to observable motion.

Is Momentum a Force?

Is Momentum, then, simply another word for force?

Again, the two should not be identified.

In conventional physics, force is a specific physical quantity used to describe changes in motion. It allows acceleration to be calculated and interactions to be represented quantitatively with extraordinary precision.

DISTANCE does not attempt to add another kind of force.

It does not claim that a new “Momentum force” acts instead of gravity.

Nor does it attempt to replace gravitational, electromagnetic, or nuclear interactions with a single force called Momentum.

Momentum is not the name of a particular interaction.

At a more general level, it refers to:

the directionality of change expressed by a Difference when that Difference is effective within a relationship.

A force may quantitatively describe the change of motion in a particular physical situation. Momentum in DISTANCE is a concept for interpreting the relational directionality through which such change becomes manifest.

The two may encounter the same physical phenomenon at different explanatory levels, but they are not identical concepts.

Without this distinction, DISTANCE would amount to little more than renaming the vocabulary of conventional physics.

Its purpose is not to replace names.

It is to reconsider the starting point of explanation.

Whose Property Is Momentum?

Another familiar intuition now needs to be examined.

We commonly say:

“This object has momentum.”

Within conventional mechanics, this is entirely natural.

There is an object, the object has mass and velocity, and its momentum can therefore be calculated.

DISTANCE does not begin by treating Momentum exclusively as a property stored inside an independently existing object.

If Difference is relational and Distance is relational, then Momentum—the directionality emerging within that relationship—must also be relational.

Suppose A exists entirely alone.

To say that A contains some absolute Momentum would immediately raise the question: direction relative to what?

Direction requires relationship.

Change also requires distinguishability between states.

Momentum is therefore not a tiny arrow stored inside an isolated object.

There may be one directionality within the A-B relationship.

There may be another within the A-C relationship.

There may be yet another within the A-D relationship.

This matters because:

one object can participate simultaneously in multiple Momentum relationships.

What we casually call “the Momentum of A” may therefore be an observational summary centered on A, combining the consequences of multiple relationships.

At the relational level, there need not be only one Momentum.

Countless Momentum may coexist.

One Object, Many Directions

Suppose A is simultaneously related to B, C, and D.

There is one Difference in the A-B relationship.

There is another in A-C.

There is yet another in A-D.

Each Difference exists under different relational conditions.

There is therefore no reason to assume that the directionality associated with every relationship must be the same.

The A-B relationship may tend to change A in one direction.

The A-C relationship may have another directionality.

A-D may have yet another.

These directions coexist.

The change we actually observe in A need not reproduce any one relationship in isolation.

Several directionalities may reinforce one another.

They may counteract one another.

One direction may be constrained by another relationship.

Another may become comparatively more visible.

Observable change may therefore be understood not as the simple consequence of

one isolated relation but as an effective directionality emerging from the present balance among multiple coexisting Momentum.

At this point, we do not need a mathematical definition of that balance.

What matters is the conceptual shift.

Instead of assuming that one object possesses one independent Momentum, we allow the possibility that:

one observable state may be the result of many Momentum relationships.

Effective Momentum

If many Momentum coexist, which direction do we actually observe?

Not every directionality appears equally.

One relational direction may be almost completely counteracted by another.

Some Momentum may operate primarily within the internal relationships that maintain a configuration and therefore appear only weakly as external motion.

Another Momentum may become comparatively dominant and appear macroscopically as motion or another observable change.

The directionality that actually becomes manifest under the present relational conditions can be called effective Momentum.

The word effective does not imply that the other Momentum do not exist.

It implies the opposite.

Multiple Momentum coexist, but under a particular relational configuration, some resulting directionality becomes observable.

This also changes how we can think about the distinction between rest and motion.

We need not divide the world into states that possess Momentum and states that do not.

Both may contain numerous Momentum relationships.

In one case, an effective Momentum may become strongly visible as macroscopic displacement.

In another, multiple directionalities may counterbalance or constrain one another internally, so that displacement is not observed.

Potential Momentum

We can now consider directionality that has not yet become sufficiently visible as observable change.

A bow is drawn and held.

The arrow remains still.

At the macroscopic level, there is no motion.

Yet when the string is released, the arrow moves in a particular direction.

That direction did not suddenly appear without cause at the moment of release.

The relationship among the bow, string, and arrow already contained a state capable of producing that change.

Similarly, an object may remain stationary at a height until some constraint changes.

A pressure Difference may exist while a closed valve prevents flow.

In each case, a relationship contains directionality of possible change, while other constraints prevent it from becoming sufficiently manifest as observable motion.

DISTANCE distinguishes such a condition using the expression Potential Momentum.

Potential Momentum should not be understood as a new form of physical energy.

Nor is it intended simply to replace the conventional concept of potential energy with another name.

It refers, at the level of explanation, to:

a directionality of change that is already contained in a relationship but has not yet become sufficiently manifest as observable change.

The purpose of the expression is simple.

It separates:

the existence of Momentum

from

the existence of observable motion.

Directionality can exist before motion becomes visible.

When relational constraints change, that directionality may become effective and appear as motion or some other form of change.

Momentum Is Not a Substance That Is Stored and Moved

The word Momentum can easily tempt us to imagine a quantity stored somewhere and then transferred from one place to another.

That is not how Momentum is defined in DISTANCE.

It is not a tiny substance or fluid.

It is not an entity stored in a fixed amount inside one object and later delivered to another.

Momentum is directionality.

Its magnitude and direction may therefore change as relationships change.

If a new relationship becomes effective, a Momentum that was previously almost invisible may become important.

If another relationship changes, a previously dominant directionality may weaken.

As the relational network changes, effective Momentum may also be continually re-configured.

In this sense, Momentum in DISTANCE is better understood not as a fixed possession but as directionality continuously reconstituted through relational states.

This will later become important when we reinterpret physical motion at larger scales.

Momentum Is Always Present

Following the argument this far leads to an important proposition.

Countless Differences exist throughout the universe.

Among them are Differences that are effective within actual relationships.

If such relationships contain directionality of change, it becomes difficult to divide reality simply into conditions with Momentum and conditions without it.

Relationships remain even when observable motion is absent.

Differences remain.

Multiple directionalities may constrain or counterbalance one another.

DISTANCE therefore states:

The existence of Momentum is never the question. Momentum is always present.

This sentence is not intended as a mathematical declaration that the exact magnitude of Momentum can never take a value of zero under any formal description.

Its more important meaning is relational.

The fact that we observe an object at rest does not justify the conclusion that every directionality of change associated with that object has disappeared.

Momentum is not identical to observable motion.

It is a broader relational directionality.

The Observable Universe Is a Question of Which Momentum Becomes Effective

If Momentum is always present, what should we ask?

Not:

“Is there Momentum?”

The question becomes:

Among countless coexisting Momentum, which becomes effective here and now?

A structure may contain numerous relationships.

Some Momentum may appear as internal binding.

Some may contribute to displacement of the structure as a whole.

Some may change its form.

Some may remain constrained by other directionalities.

An observable physical phenomenon therefore need not be the expression of one fundamental Momentum acting alone.

The continuously changing balance among countless Momentum may determine which directionality becomes effective at a given moment.

This can be stated more precisely:

The existence of Momentum is never the question. Momentum is always present. The observable universe is simply the present expression of which Momentum becomes effective.

This proposition becomes one of the central principles of the DISTANCE interpretation of motion.

Are Independent Momentum Acting on Independent Objects?

We can now move one step further.

We often imagine the physical world in the following way.

First, there are objects.

Each object exists independently.

Forces act among those objects.

The objects then acquire momentum and move.

This picture is extraordinarily useful for observation and calculation.

But the relational argument developed so far allows another possibility.

Difference is relational.

Distance is relational.

Momentum is relational.

Why, then, must objects and Momentum be assumed from the beginning to belong to entirely separate explanatory categories?

Countless Momentum relationships may form a particular configuration, and part of that configuration may appear at our observational scale as what we call an object.

In that case, an object would not merely be a finished container that possesses Momentum.

It could instead be understood as an observable configuration in which Momentum relationships are organized into a particular state.

We do not need to establish that conclusion here.

The problem of structures and Islands will be examined later.

But at this stage we can at least say:

The universe need not be understood only as a collection of independent objects acted upon by independent Momentum.

More strongly:

The universe is therefore never composed of independent objects acted upon by independent Momentum. It is composed of countless coexisting Momentum whose continuously changing balance determines the observable directionality of every physical system.

This is not yet a theory of how those balances are resolved.

It is a statement about the relational architecture within which observable directionality emerges.

Momentum Does Not Have Only One Direction

From this perspective, even the phrase “the direction of motion” needs to be treated carefully.

Suppose an object is moving to the right.

At the level of observable mechanics, it is natural to say that its momentum points to the right.

That description is valid.

But at a deeper relational level, the object need not participate only in rightward relationships.

Another relationship may contain a leftward Momentum.

Internal binding relations may contain still other directionalities.

Some Momentum may operate with almost no direct expression in the macroscopic displacement of the object.

The observed rightward motion does not mean that all these other directionalities have disappeared and only one remains.

It may mean that, within the present relational balance, a rightward directionality has become effective.

This distinction later becomes important when considering linear motion and rotation.

Even an object that appears to move along a straight line need not be a simple system containing only one Momentum.

It may instead be a system in which multiple Momentum maintain a sufficiently stable effective direction.

But that problem belongs to a later chapter.

What Lies Between Momentum and Distance?

At this point, a natural question arises.

Distance expresses the effective nearness or farness of a relationship.

Momentum expresses the directionality of change that becomes manifest through that relationship.

What, then, is the exact relation between them?

Does a closer relationship imply a larger Momentum?

What happens to Distance when Momentum becomes weaker?

Are the two simply proportional, inversely related, or connected in some more complex way?

These questions are essential.

But Section 1.5 stops here.

The task of this section has been to establish what Momentum means and what role it plays.

Difference describes relational non-equivalence.

Distance expresses the effective nearness or farness of that relationship.

Momentum expresses the directionality of change associated with an effective relationship.

Before we formally connect Momentum and Distance, however, one conceptual distinction must be secured once more.

Why is a large Difference not the same thing as a short Distance?

Why can similar Differences exist at different Distances?

And why must this distinction remain stable if the relation between Momentum and Distance is not to be reversed?

The next section therefore returns briefly to these two concepts and separates their roles more precisely.

1.6 Difference and Distance

Only after that distinction has been secured can we formally examine Momentum–Distance Duality.

6 Difference and Distance

Following the discussion so far, Difference and Distance may appear to be very closely related concepts.

Difference exists.

That Difference becomes effective within a relationship.

And we have called the effective nearness or farness of that relationship Distance.

From this sequence alone, it is tempting to make a simple correspondence: a larger Difference means a shorter Distance, while a smaller Difference means a longer Distance.

But this immediately creates a problem.

Difference and Distance are closely connected, but they are not two different names for the same thing.

This distinction is more than a matter of terminology.

It becomes a critical reference point for the entire direction of the argument when we later discuss Momentum and Equalization.

If Difference and Distance are confused, the resolution of Difference can easily be misread as a reduction of Distance.

Once that happens, the direction of everything that follows can be reversed.

We therefore need to pause here and separate the two concepts once more before moving forward.

Difference Asks “How Different?”

In its simplest sense, Difference is the fact that two states are not equivalent.

A and B have different temperatures.

Their pressures differ.

Their electrical potentials differ.

Their compositions differ.

A previous state differs from a later state.

Difference refers to this non-equivalence.

In a physical relationship, such Difference may be expressed as an Energy Gap.

But Difference itself does not directly tell us the strength of the relationship.

Suppose, for example, that there are two situations with the same temperature difference.

There is a temperature difference of twenty degrees between A and B.

There is also a temperature difference of twenty degrees between C and D.

If we consider only the magnitude of Difference, the two cases appear identical.

But suppose A and B can exchange heat very easily, while C and D can exchange almost none.

The significance of the two Differences within their actual relationships is then not the same.

The same Difference exists, but the degree to which that Difference is effective within the present relationship is different.

Difference alone does not tell us this.

Difference tells us only how non-equivalent the states are.

Distance Asks “How Effective Is That Difference?”

Distance enters precisely at this second question.

The fact that A and B are different and the degree to which that Difference is effective in their actual relationship are separate matters.

DISTANCE therefore asks:

How effective is that Difference within the present relationship?

This question does not merely restate the amount of Difference.

It includes the conditions of the relationship.

The same Difference can correspond to different Distances when the relational conditions differ.

Conversely, Differences of different magnitudes may have similar relational effectiveness.

We therefore cannot assume a one-to-one correspondence between Difference and Distance.

The distinction can be stated simply:

Difference describes the degree of non-equivalence. Distance describes the effectiveness of that non-equivalence within a relationship.

The two concepts are connected, but neither can replace the other.

Does a Large Difference Necessarily Mean a Short Distance?

Intuitively, it is easy to assume that a larger Difference should produce a stronger relationship.

A larger temperature difference may be associated with more rapid thermal change.

A larger pressure difference may produce stronger flow.

A larger potential difference may make greater change possible.

It is therefore tempting to write immediately:

larger Difference $\rightarrow$ shorter Distance

But this is too quick a simplification.

The magnitude of Difference alone cannot determine the full effectiveness of a relationship.

No matter how large a Difference may be, if the relevant relationship is effectively blocked, that Difference may have little significance for the change currently taking place.

Conversely, a relatively small Difference may play an important role in actual change if the relational conditions make it strongly effective.

Distance therefore cannot immediately be defined as a simple function of the magnitude of Difference.

Distance must include both Difference and the conditions of relationship.

This is crucial.

When we later speak of a “short Distance,” it is not simply another way of saying “a large Difference.”

It means that the Difference is strongly effective within the present relationship.

Energy Gap Is Not Distance Either

The same problem arises with Energy Gap.

Earlier, we used Energy Gap to describe cases in which Difference is expressed energetically within a physical relationship.

Could we then simply define a large Energy Gap as a short Distance?

Again, caution is necessary.

Energy Gap represents energetic non-equivalence.

Distance represents how effective that energetic non-equivalence is within the present relationship.

Energy Gap and Distance are therefore not identical concepts.

A large Energy Gap may exist while a relationship is constrained in such a way that the Gap does not immediately appear as strong change.

A smaller Energy Gap may become highly significant within a relationship that is especially responsive to it.

Thus:

Energy Gap $\neq$ Distance

Nor can we simply declare at this stage that:

Distance = the inverse of Energy Gap.

A more specific mathematical relationship may eventually be explored within particular physical formulations.

But conceptual separation must come first.

Energy Gap tells us what is energetically non-equivalent and by how much; Distance tells us how effective that non-equivalence is within a relationship.

Difference Can Be Resolved

Difference can change through a process of transformation.

A temperature difference decreases.

A pressure difference changes.

A potential difference changes.

Differences in composition may change.

When one state changes into another, the Difference that previously existed between them may also change.

In this sense, Difference can be resolved.

The word resolution is important here.

A reduction in Difference means that the non-equivalence between states has decreased.

Does Distance therefore decrease as well?

This is exactly where the two concepts are easily confused.

Intuitively, when two states become less different, we may want to say that they have “become closer.”

But this mixes the language of Difference with the language of Distance.

A smaller Difference means that the states have become more similar.

A short Distance means that their relationship is more strongly effective.

These are not the same kind of nearness.

The reduction of Difference must therefore not be automatically translated into a reduction of Distance.

If this distinction is lost, the direction of Equalization later becomes completely reversed.

“Becoming More Similar” Is Not the Same as “Becoming Closer”

The same confusion appears easily in everyday language.

Suppose two people narrow a disagreement.

We may say that “they have become closer.”

Here, closer means that their positions have become more similar.

But in the relational Distance of DISTANCE, closeness has a different meaning.

It refers to the effectiveness of the relationship.

Two states becoming more similar and their relationship becoming more strongly effective are therefore not the same phenomenon.

Indeed, if a particular Difference is resolved so that the two states no longer strongly tend to change one another through that Difference, the effectiveness of that relationship may weaken.

The states may have become more similar while their relational Distance becomes longer.

At first this may sound paradoxical.

Once Difference and Distance are separated, however, it is not.

Difference is non-equivalence between states.

Distance is the effectiveness of the relationship generated through that Difference.

As non-equivalence is resolved, the relational tension associated with that Difference may also weaken.

“Becoming more similar” and “becoming relationally closer” should therefore not be assumed to point in the same direction.

Geometric Nearness Is Yet Another Question

The situation becomes even easier to confuse when geometric distance is added.

We may now have three different concepts:

First,

the magnitude of Difference

Second,

relational Distance

Third,

geometric distance

These concepts may be related, but they are not identical.

Two objects may become geometrically closer.

During that process, a particular Difference may decrease.

Or it may increase.

The effectiveness of their relationship may also change.

There is no reason to assume that all three quantities must always change in the same direction.

Whenever DISTANCE later uses the expression:

“becoming closer,”

we must therefore ask what exactly is becoming closer.

Is geometric distance decreasing?

Is Difference becoming smaller?

Or is relational Distance becoming shorter?

If these three are not distinguished, the logic can easily be reversed.

This becomes especially important when we later discuss gravity, rotation, and centripetal and centrifugal behavior.

Motion in a geometrically inward direction does not by itself mean that energy/effective Distance is decreasing.

Conversely, geometrically outward motion does not automatically mean that relational Distance is increasing.

We do not need to analyze those forms of motion here.

But the principle that these different forms of “distance” must not be conflated should be established now.

A Reduction of Difference May Mean a Weakening of Relationship

Consider only one simple relationship.

There is a particular Difference between A and B.

Because of that Difference, A and B strongly affect one another.

Over time, the Difference becomes smaller.

There is then less reason for that particular Difference to continue sustaining the relationship with the same effectiveness as before.

The reduction of Difference may therefore be associated with a reduction in the effectiveness of the relationship based on that Difference.

This point is important for understanding the direction of DISTANCE intuitively.

We often imagine stabilization or resolution as a process of “coming closer.”

But in DISTANCE, as a particular Difference is resolved, the relationship organized around that Difference may become weaker.

A resolved relationship may therefore become relationally more distant.

The later sections will describe this more precisely.

For now, one point is enough:

A decrease in Difference does not imply a decrease in Distance.

Distance Can Continue to Change Between the Same Objects

Even when A and B remain the same identifiable objects, their Distance need not remain constant.

Relational conditions can change.

A changes.

B changes.

The surrounding environment changes.

Other relationships that mediate or constrain the A-B relationship also change.

The effectiveness of the Difference between the same two objects may therefore continue to vary.

Distance is not a fixed property.

The fact that the labels A and B remain the same does not mean that their relational Distance remains the same.

This also distinguishes relational Distance from geometric distance.

Two objects may remain at almost the same geometric positions while their relational Distance changes substantially.

Conversely, their geometric positions may change considerably while the effectiveness of a particular relationship remains relatively persistent.

Distance is therefore more directly associated with the state of a relationship than with the identity of the objects participating in it.

This later becomes important when we consider Islands and Structures.

The persistence of a Structure does not imply that all internal Distances remain fixed.

Countless relationships may continue to change while a relatively persistent overall configuration remains observable.

That issue, however, belongs later.

Can Distance Exist Without Difference?

We can also ask the reverse question.

If there is no Difference at all, does Distance still have meaning?

Suppose A and B are completely equivalent in every relevant respect and no non-equivalence exists between them.

What would it mean to describe the relational effectiveness of a Difference that does not exist?

If there is no particular Difference, we cannot ask how effective that Difference is.

In this sense, Distance is not completely independent of Difference.

Distance presupposes Difference.

But it is not identical to Difference.

Their relation can be expressed as follows:

Difference is a condition for relational Distance to have meaning. Distance describes the effectiveness of Difference within a relationship.

The two concepts are therefore hierarchically connected without being interchangeable.

Can Difference Alone Fully Describe a Relationship?

Conversely, can Difference alone explain actual relationships?

As we have already seen, it cannot.

The same Difference may have very different effectiveness under different relational conditions.

With Difference alone, we may be able to say:

“these two states differ by this much.”

But we cannot fully explain:

“how strongly does that Difference matter in this relationship now?”

Difference and Distance therefore answer different questions.

Difference describes non-equivalence between states.

Distance describes the effectiveness of that non-equivalence within a relationship.

Only by keeping these questions separate can the directionality of later arguments remain clear.

Difference, Distance, and the State of a Relationship

We can now place the two concepts together.

There is a Difference between A and B.

That Difference exists within certain relational conditions.

The relationship may be open or constrained.

Other relationships may reinforce it or limit it.

Within all these conditions, the Difference becomes effective to some degree.

DISTANCE expresses the nearness or farness of that effective relationship as Distance.

Distance is therefore not determined by Difference alone.

The entire relational state matters.

This distinguishes DISTANCE from a simple difference-based theory.

The argument is not merely that larger differences always produce stronger change.

We must also ask:

Within what relational configuration does a particular Difference exist?

What, Then, Determines Actual Change?

At this point, another question naturally follows.

We now understand why Difference and Distance must be distinguished.

But how are the direction and strength of actual change determined?

In the previous section, we introduced Momentum as the directionality of change expressed within a relationship.

There is Difference.

That Difference has a certain effectiveness within a relationship.

And that relationship exhibits directionality.

The next task is to connect these concepts.

The crucial question is:

What does a shorter Distance imply about Momentum?

And conversely:

What happens to Distance as Momentum is resolved?

These questions go beyond terminology.

They determine the basic dynamics of DISTANCE.

But if we simplify the relationship too quickly, Difference and Distance may become entangled once again.

Section 1.6 has therefore done something more fundamental first: it has secured the boundary between them.

The Two Concepts in Their Simplest Form

The purpose of this section has not been to introduce another new concept.

It has been to prevent two existing concepts from collapsing into one.

In summary:

Difference Non-equivalence between states. “What is different, and by how much?”

Distance The effective nearness or farness of that Difference within a particular relationship. “How effective is that Difference in this relationship?”

The distinction must remain intact.

A decrease in Difference does not imply a decrease in Distance.

An increase in Difference does not automatically imply a shorter Distance.

A larger Energy Gap does not by itself mean a shorter Distance.

Relational conditions must also be considered.

Geometric distance belongs to yet another level of observation.

These different meanings of nearness must not be collapsed into one.

The Next Step — The Relationship Between Momentum and Distance

Only now can we move to the next question.

Momentum is the directionality of change.

Distance is the effective nearness or farness of a relationship.

How, then, are the two related?

When a relationship is closer and more strongly effective, does Momentum become larger?

As Momentum weakens, does Distance become longer?

Why does their relationship appear to take the form of an inverse correspondence?

And in what sense can this relationship be called a duality?

These are the questions of the next section:

1.7 Momentum–Distance Duality

The reason Section 1.6 has separated Difference from Distance so carefully is precisely to make that relationship possible to state without ambiguity.

If Difference and Distance are confused, the relationship between Momentum and Distance can be reversed as well.

With that distinction now secured, we can examine their correspondence directly.

7 Momentum–Distance Duality

In the preceding sections, we distinguished three concepts.

Difference is non-equivalence between states.

Distance expresses how effective that Difference is within a particular relationship, in terms of relational nearness or farness.

And Momentum is the directionality through which that effective relationship becomes manifest in actual change.

We separated these concepts not because they should remain independent, but because we now need to connect them without confusing their roles.

In particular, Distance and Momentum have a relationship that runs through the entire framework of DISTANCE.

The more strongly effective a relationship is, the shorter we describe its Distance as being.

And the more strongly the directionality of change becomes effective within that relationship, the greater its effective Momentum can be said to be.

Conceptually, this gives us the following correspondence:

shorter Distance $\leftrightarrow$ stronger effective Momentum

and conversely,

longer Distance $\leftrightarrow$ weaker effective Momentum

DISTANCE calls this correspondence Momentum–Distance Duality.

This, however, is not merely a declaration of a mathematical inverse proportionality.

The more important point is that Distance and Momentum allow us to read the same relational state from two different perspectives.

Distance describes the relationship in terms of nearness and farness.

Momentum describes the same relationship in terms of directionality of change.

They are not two independent substances.

They are two aspects through which one relational state becomes intelligible.

A Close Relationship Is a Strongly Effective Relationship

In Section 1.4, we distinguished Distance from geometric separation.

Two objects being geometrically close does not necessarily mean that they are also close in the sense used by DISTANCE.

Conversely, geometric separation alone does not necessarily determine how distant a particular relationship must be.

In DISTANCE, a close relationship is one that is strongly effective.

Suppose the A-B relationship is more strongly connected to changes in A than the A-C relationship.

With respect to those relationships, we can describe the Distance between A and B as shorter.

What matters is how strongly the relationship participates in actual change.

And change has directionality.

That directionality is Momentum.

Distance and Momentum are therefore not two concepts that happen to correlate.

If Distance expresses the effectiveness of a relationship, and Momentum expresses the directionality of that effective relationship, then both concepts refer from the beginning to different aspects of the same relational state.

The more strongly effective the relationship becomes, the stronger its Momentum may become.

This is why shorter Distance corresponds to stronger effective Momentum.

This Does Not Mean “Large Difference = Large Momentum”

Here the distinction established in Section 1.6 becomes essential again.

We must not simplify the argument into:

large Difference $\rightarrow$ large Momentum

The magnitude of Difference alone does not determine how effective Momentum will be.

A very large Difference may exist within a strongly constrained relationship and therefore have little effectiveness in the change currently taking place.

Conversely, a relatively small Difference may produce a strong directionality of change if the relationship is highly responsive and effective.

What matters in Momentum–Distance Duality is therefore not the absolute magnitude of Difference.

It is how effective that Difference is within the present relationship.

Without this distinction, Difference, Distance, and Momentum would once again collapse into what appears to be a single quantity.

DISTANCE does not claim:

A large Difference necessarily means a short Distance.

Nor does it claim:

A large Difference necessarily means a large Momentum.

But when a Difference is strongly effective within a particular relationship, that relationship can be described as having a shorter Distance, and its directionality of change may appear as stronger effective Momentum.

Momentum Is Not Resolved by Decreasing Distance

We now arrive at the most important issue of directionality.

In everyday language, we often describe the resolution of tension or difference as a process in which two states “come closer.”

A temperature difference decreases.

A pressure difference decreases.

Two different states become more similar.

This makes it easy to reason as follows:

Difference is resolved $\rightarrow$ the states become closer $\rightarrow$ Distance decreases.

Within DISTANCE, however, this is the wrong connection.

Here, “becoming closer” has been used to mean that Difference decreases and the states become more similar.

But Distance in DISTANCE does not measure similarity between states.

It describes how strongly a Difference is effective within their relationship.

As a particular Difference is resolved, the Momentum based on that Difference weakens.

As Momentum weakens, the relationship becomes less effective than before.

Its Distance therefore becomes not shorter, but longer.

The direction is thus:

Momentum resolution $\rightarrow$ increasing Distance

This direction must remain fixed throughout the arguments that follow.

Resolution Does Not Mean the Disappearance of Relationship

If Momentum is resolved and Distance increases, this does not mean that every relationship between two objects disappears.

A and B may participate in more than one relationship.

Momentum based on one Difference may weaken.

Another Difference may become important.

While one A-B relationship becomes more distant, another relationship between them

may become closer.

And when A changes, its relationships with C, D, and other states change as well.

The actual world is therefore not a simple process in which one Momentum is created, acts, and then vanishes completely before anything else happens.

The relational network is continuously reconfigured.

Some Momentum are resolved.

Others become effective.

While one short Distance becomes longer, another short Distance may emerge.

Change in DISTANCE should therefore not be understood as a process in which all relationships simply become more distant in one uniform direction.

It is better understood as a continuous rearrangement of the relational network through the resolution and reconfiguration of Momentum.

Why Call It a “Duality”?

Why describe the relationship between Distance and Momentum as a duality rather than simply a correlation?

Because they are not merely two independent quantities that happen to vary together.

Consider a relationship that is strongly effective.

In the language of Distance, we describe it as close.

In the language of Momentum, we say that a strong directionality of change is effective within it.

These are not two unrelated phenomena appearing at the same time.

They are two descriptions of the same relational state.

Conversely, when the Momentum of that relationship is resolved and its effectiveness weakens, the same change can be read in two ways.

From the perspective of Momentum, its directionality weakens.

From the perspective of Distance, the relationship becomes more distant.

One concept therefore describes the effective separation of a relational state, while the other describes its effective directionality.

In this sense, the two are dual:

Distance is the distance-like expression of a relational state, while Momentum is the directional expression of the same relational state.

How Far Can We Take the Language of an Inverse Function?

We may describe the relationship between Momentum and Distance as an inverse relation.

The larger the Momentum, the shorter the Distance.

The smaller the Momentum, the longer the Distance.

Conceptually, this is useful.

But we should not immediately declare a universal mathematical law such as

$$M = k \overline{D}.$$

Before such an equation could be justified, we would need quantitative definitions of both Momentum and Distance.

Different types of physical relationship might also produce different functional forms.

The claim at this stage is therefore conceptual rather than a universal mathematical law.

We can express the directional correspondence as:

$$D \downarrow \iff M_{\text{eff}} \uparrow$$

and

$$M_{\text{eff}} \downarrow \iff D \uparrow.$$

Here, D means relational Distance, not geometric distance.

Likewise, M_{eff} refers to effective Momentum in the sense developed here, not simply to the conventional

This limitation must remain explicit.

Momentum Resolution Is an Increase in Distance

We can now state the central proposition of this section more directly.

A relationship having a short Distance means that its Momentum is strongly effective.

When that Momentum is resolved through change, the effectiveness of the relationship decreases.

Distance therefore increases.

In compressed form:

Momentum is resolved as Distance increases.

This sentence captures a fundamental direction in DISTANCE.

Momentum is not resolved because two states move toward one another in energy Distance.

The direction is the opposite.

As a strongly effective relationship is resolved, its relational Distance becomes longer.

This distinction becomes particularly important when we later reinterpret physical motion.

Even when an object is observed moving geometrically toward another object, we cannot infer from that fact alone that its energy or relational Distance is decreasing.

The direction of geometric motion and the direction of relational Distance are different questions.

Geometric Inward Motion Is Not Distance Decrease

This distinction deserves to be made explicit.

When we think about a circle or sphere, we naturally distinguish between the center and the outside.

Motion toward the center feels like “coming closer,” while motion outward feels like “moving farther away.”

But this is a description within geometric coordinates.

Energy or relational Distance in DISTANCE does not begin with an intrinsic distinction between “inward” and “outward” in that geometric sense.

If a relationship produces motion that appears geometrically inward, that does not mean the Distance of that relationship is decreasing.

If the motion is part of the resolution of Momentum, its relational Distance is still changing in the direction of increase.

Likewise, observing geometrically outward motion does not by itself establish a separate mechanism of Distance increase.

Both may be observable expressions of different Momentum relationships undergoing resolution.

The following identifications must therefore be rejected:

geometric inward = Distance decrease

and

geometric outward = Distance increase

Geometric direction and the direction of relational Distance must remain separate.

This principle later becomes crucial when DISTANCE reinterprets gravity, circular motion, and the relation between centripetal and centrifugal behavior.

Potential Momentum Follows the Same Relationship

In Section 1.5, we introduced Potential Momentum as directionality that exists within

a relationship but has not yet become sufficiently manifest as observable motion.

Potential Momentum does not lie outside Momentum–Distance Duality.

Consider a strongly effective relationship whose observable change is prevented by another constraint.

The relationship may still possess a short Distance.

And strong Potential Momentum may still exist within it.

If the constraint is removed, that Momentum may become effective as observable change.

Whether macroscopic motion is currently visible therefore does not determine whether Momentum–Distance Duality applies.

What matters is how effective the relationship is.

This reinforces once more the distinction:

Momentum $\neq$ observable motion

Resolving One Relationship Changes the Momentum of Others

In reality, there is never only an isolated A-B relationship.

A also participates in relationships with C, D, E, and countless other states.

B does the same.

When the Momentum of A-B is resolved and its Distance increases, the states of A and B change.

As a consequence, their Differences and Distances with other states may change as well.

An increase in A-B Distance may make A-C Distance shorter.

A-D Momentum may become newly effective.

A relationship that previously mattered very little may become a new preferred relation.

This matters because the universe does not resolve one relationship completely and only then proceed to the next.

Countless relationships change simultaneously.

The universe therefore does not simply move along one Momentum.

It can instead be understood as:

a process in which countless Momentum continually alter one another's relational conditions and thereby reconstitute effective Momentum.

This reconnects us with the proposition introduced in Section 1.5:

Momentum is always present.

The resolution of one Momentum does not mean that Momentum itself disappears from the universe.

The relational network changes, and other Momentum become effective.

Short Distance Is Not the Goal

Another important misunderstanding must be prevented here.

Because a more strongly effective relationship has a shorter Distance, it may seem as though the universe somehow seeks shorter Distance.

The risk of this misunderstanding becomes even greater when we later use words such as preferred.

But short Distance is not the target state.

A short Distance instead represents a relational condition in which strong Momentum remains unresolved.

It indicates that a strong directionality of change is still effective.

Thus:

preferred relationship $\rightarrow$ short Distance

must not be read as:

the universe seeks shorter Distance.

Its meaning is closer to:

shorter-Distance relationships carry stronger effective Momentum and therefore become more prominent in the ongoing process of change.

Short Distance is not the endpoint of Equalization.

It is a condition under which change can become strongly effective.

“Minimum Distance” Must Be Read the Same Way

For the same reason, expressions such as minimum Distance require care.

If we say that a process “prefers minimum energy Distance,” it can sound as though minimizing Distance itself were the final purpose of the universe.

Under the definitions established here, that interpretation would be incorrect.

If one possible relationship has a shorter Distance than another, it means that its Momentum is more strongly effective.

Therefore, if an expression such as

preferred Equalization $\rightarrow$ shorter/minimum energy Distance

is used later, it must not mean:

Equalization proceeds by reducing Distance.

It means instead:

among currently available relationships, those with shorter Distance possess stronger effective Momentum and may therefore appear as preferred paths in the process of Equalization.

This distinction is essential.

If it is lost, the direction of the entire framework is reversed once again.

The Starting Condition and the Direction of Resolution Must Be Distinguished

We must therefore distinguish between:

the condition that makes one relationship more effective in change

and

the direction in which that relationship proceeds as its Momentum is resolved.

At the starting condition:

shorter Distance $\rightarrow$ stronger effective Momentum

so a shorter-Distance relationship may become more prominent in change.

But as that Momentum is actually resolved:

Momentum resolution $\rightarrow$ increasing Distance

These statements are not contradictory.

The first tells us which relationship is more strongly effective.

The second tells us how that effective relationship changes through resolution.

In compressed form:

Short Distance is the condition of strong Momentum; long Distance is the state in which more of that Momentum has been resolved.

Keeping this distinction explicit prevents terms such as preferred, minimum, resolution, and Equalization from becoming conflated.

One Relationship Becomes More Distant, but the World Does Not Simply Disperse

If Momentum resolution means increasing Distance, does that imply that everything in the universe simply moves farther and farther apart?

No.

While the Distance of one relationship increases, the Distance of another may become shorter.

As the A-B relationship weakens, A-C may become more strongly effective.

An existing structure may dissolve while another structure forms.

The resolution of one Momentum may generate or expose new Differences and new Momentum.

Increasing relational Distance therefore does not imply the simple geometric dispersion of all structure.

The broader process is one of relational rearrangement.

Some relationships become more distant.

Others become newly close.

Some Momentum weaken.

Others become stronger.

And the structures and motions we observe change according to which Momentum become effective at a given moment.

This is the starting point for later interpreting the universe not as a static collection of objects, but as a continuously reconfigured network of Momentum relationships.

Momentum–Distance Duality in One Sentence

We can now compress the central idea of this section into one proposition:

Momentum is the observable directionality associated with unresolved Distance.

Where relational Distance remains strongly unresolved, Momentum is effective.

The more strongly effective the relationship, the shorter its Distance and the stronger its effective Momentum.

As Momentum is resolved, the relationship weakens and Distance increases.

Thus,

shorter Distance $\iff$ *stronger effective Momentum*,

while

Momentum resolution $\implies$ *increasing Distance*.

This is the central meaning of Momentum–Distance Duality in DISTANCE.

The Next Step — Equalization

One question remains.

Why is Momentum resolved?

And within a relational network containing countless coexisting Momentum, what overall direction does that resolution take?

If one Momentum is resolved while another becomes effective, and one Distance increases while another relationship becomes shorter, is there any common pattern underlying the entire process?

DISTANCE addresses this question through the concept of Equalization.

The direction must now remain unambiguous.

Equalization is not a process that reduces Distance until everything converges into one maximally close state.

It begins with Difference and Energy Gap.

Momentum appears through effective relationships.

And through the process by which that Momentum is resolved, Distance increases.

The conceptual sequence established so far is therefore:

Difference $\rightarrow$ Distance $\rightarrow$ Momentum $\rightarrow$ Equalization $\rightarrow$ Momentum resolution and increasing Distance

What remains is to explain exactly what Equalization means, and why it should not be confused with simple homogenization, convergence, or rest.

That is the subject of the final section of this chapter:

1.8 Equalization — The Continuing Direction of Change.

8 Equalization — The Continuing Direction of Change

So far, we have begun from the smallest conceptual units of change.

When two states are not equivalent, there is Difference.

Within a physical relationship, that Difference may appear as an Energy Gap.

But not every Difference is equally effective in producing actual change. Some Differences form strongly effective relationships, while others produce almost no observable change. To express this relational effectiveness in terms of nearness and farness, we introduced the concept of Distance.

And when Difference becomes effective within a relationship, the relationship does not merely remain in a state of being “strong.” Actual change exhibits directionality.

We called that directionality Momentum.

In the previous section, we clarified the relationship between Momentum and Distance.

A short Distance means a strongly effective relationship.

Such a relationship may exhibit stronger effective Momentum.

And as that Momentum is resolved, the effectiveness of the relationship weakens and Distance increases.

One final question therefore remains.

Why does this change continue?

When one Momentum is resolved, does change end?

Or does that change alter the conditions of other Differences, Distances, and Momentum, thereby generating the conditions for further change?

DISTANCE calls this continuing direction of change Equalization.

Does Equalization Make Everything the Same?

The word Equalization most easily suggests homogenization.

Hot water and cold water mix.

Solutions of different concentrations diffuse.

Regions of different pressure become connected.

As time passes, Difference decreases and the states become more similar.

It is therefore tempting to think of Equalization simply as:

Difference $\rightarrow$ zero

This may describe one aspect of Equalization.

But it is not sufficient to describe Equalization as a whole in DISTANCE.

In the actual world, the reduction of one Difference does not mean that all Differences decrease simultaneously.

If the Difference between A and B changes, the state of A changes.

Then the Difference between A and C also changes.

The A-D relationship changes as well.

B also participates in countless other relationships, so a change in B alters still other Differences.

The resolution of one Difference is therefore not an isolated event.

It simultaneously changes other parts of the relational network.

Equalization should thus be understood not simply as a process that drives every Difference toward zero, but as:

a process in which Momentum generated by effective Difference is resolved while the entire relational network is continuously reconfigured.

The Metaphor of Smoothing — and Its Limits

The idea of smoothing can be intuitively useful when explaining Equalization.
An uneven surface gradually becomes flatter.
Differences in height diminish.
Energetic inequalities are reduced.
Such images help us understand one direction of change.
But problems arise if the metaphor is taken literally.
The universe is not a single fixed surface.
While one region is being equalized, the relational network itself changes.
As the network changes, new Differences may appear.
Differences that were previously almost ineffective may become important.
Some relationships may become more distant while others become closer.
Equalization is therefore not like repeatedly smoothing a landscape whose shape has already been fixed.

It is closer to a process in which the landscape to be equalized is itself continually transformed by the act of Equalization.

For this reason, this book uses Equalization as the central term rather than smoothing.

Equalization includes not only the reduction of relational non-equivalence but also the continuing transformation of the relational conditions under which that non-equivalence becomes effective.

Equalization Is Not a Process of Decreasing Distance

At this point, the direction established in the previous section must be fixed once again.

When Difference decreases, we easily say that two states have “become closer.”

This makes it tempting to interpret Equalization as a process in which Distance decreases.

But that confuses Distance in DISTANCE with geometric distance or similarity.

In DISTANCE, short Distance means a strongly effective relationship.

It is a relationship in which strong Momentum remains unresolved.

When that Momentum is resolved through Equalization, the effectiveness of the relationship weakens.

Distance therefore increases.

In other words:

Equalization does not reduce Distance. Equalization resolves Momentum, and resolved

Momentum corresponds to increasing Distance.

This direction is fundamental to DISTANCE.

We must therefore not write:

Difference $\rightarrow$ Distance $\rightarrow$ Momentum $\rightarrow$ Distance decrease $\rightarrow$ Equalization

That sequence reverses the argument.

The direction established so far is:

Difference $\rightarrow$ Distance $\rightarrow$ Momentum $\rightarrow$ Equalization $\rightarrow$ Momentum resolution and increasing Distance

Equalization does not seek shorter Distance.

Shorter Distance instead characterizes a relationship in which strong Momentum is still effective.

Why, Then, Do Short-Distance Relationships Become Prominent?

An apparent contradiction now arises.

If Equalization results in increasing Distance, why does change often become stronger in relationships with shorter Distance?

Momentum–Distance Duality removes the contradiction.

A shorter Distance means a more strongly effective relationship.

Such a relationship therefore carries stronger effective Momentum.

When countless relationships coexist, those with stronger Momentum may become more prominent in actual change.

Short Distance is therefore not the destination of Equalization.

It is a condition from which Equalization becomes strongly effective.

We can thus maintain both propositions:

shorter Distance $\rightarrow$ stronger effective Momentum

and

Equalization $\rightarrow$ Momentum resolution $\rightarrow$ increasing Distance

The first describes which relationship is presently more effective in change.

The second describes the direction in which that relationship changes as its Momentum is resolved.

They do not contradict each other.

What “Preferred Equalization” Means

We can now define the expression preferred Equalization more precisely.

The word preferred does not imply that the universe knows a goal and deliberately chooses a better path.

No intention is required.

No purpose needs to be assumed.

Nor is there any need to imagine a calculated choice.

Among countless relationships, some have shorter Distance.

Those relationships carry stronger effective Momentum.

Their changes may therefore become more strongly expressed than those of other relationships.

It is in this sense that a path may be described as preferred.

Thus, if we use an expression such as:

preferred Equalization $\rightarrow$ shorter/minimum energy Distance

its meaning must be read carefully.

It does not mean:

Equalization minimizes Distance.

Rather, it means:

among currently available relationships, those with shorter Distance carry stronger effective Momentum and therefore become more prominent as effective paths of Equalization.

And as Momentum is resolved along such a path, the Distance of that relationship increases.

Thus, shorter Distance describes the condition of the relationship through which Equalization becomes preferentially effective, whereas increasing Distance describes the direction in which its Momentum is resolved.

Confusing these two would reverse the direction of DISTANCE once again.

Equalization Has No Fixed Center

Another intuition requires caution.

When we think of equilibrium or stabilization, we naturally imagine a center or final state.

Everything moves toward that point and eventually settles into a maximally stable condition.

But nothing in the definition of Equalization established so far requires such a center to be assumed in advance.

There is an A-B relationship.

There is an A-C relationship.

There is a B-C relationship.

And there are countless others.

Each contains different Difference, Distance, and Momentum.

When one Momentum is resolved, the conditions of the entire network change.

A relationship that was not previously preferred may then become preferred.

The path of Equalization is therefore not a line drawn in advance toward one final center.

As the conditions of existing relationships change, the effective paths of resolution are continually reconstituted.

Equalization therefore does not require a single fixed destination or center.

Structures that appear to contain a center may emerge later.

But we need not assume that Equalization was aiming at that center from the beginning.

A center-like configuration may instead appear as a result of repeated relational re-configuration.

We will encounter this issue again when discussing circles, spheres, rotation, and physical structures.

Resolving One Momentum Changes Others

Suppose there is strong Momentum in the relationship between A and B.

As that Momentum is resolved, the states of A and B change.

The Difference between A and C then changes as well.

The Difference between B and D may change.

The Distance of the A-E relationship may become shorter.

As a result, a previously weak Momentum may become effective.

Equalization is therefore not simply the sum of independent processes in which individual Momentum disappear one after another.

Each resolution changes the conditions for subsequent resolution.

In simplified form, we might write:

$M_1 \text{ resolution} \rightarrow \text{relational rearrangement} \rightarrow M_2 \text{ becomes effective.}$

But in the actual world, these events do not occur neatly one at a time.

Countless Momentum are simultaneously being resolved and becoming newly effective,

while continually altering one another's relational conditions.

The more accurate picture is therefore not a linear chain but a continuously changing network.

This is why Equalization can continue.

Is Equalization a Final Process That Eliminates Momentum?

If Equalization continues long enough, does all Momentum eventually disappear?

At this stage, there is no reason to conclude that it must.

The resolution of one Momentum reconfigures the relational network.

That reconfiguration may make other Differences effective.

New Distances may become relevant.

New Momentum may become effective.

The resolution of individual Momentum and the complete disappearance of Momentum are therefore not equivalent.

This returns us to the proposition established earlier:

The existence of Momentum is never the question. Momentum is always present.

Equalization is better understood not as a process that creates a universe without Momentum, but as a process that continually changes which Momentum becomes effective.

The observable universe is therefore not a fixed collection of Momentum.

Some Momentum weaken.

Others become stronger.

Some relationships dissolve.

Others form.

And the present configuration of these changes is the world we observe.

Equalization Does Not Aim at Rest

Identifying Equalization with equilibrium produces another misunderstanding.

Equilibrium is often imagined as a state in which nothing changes.

All forces cancel.

All differences disappear.

All motion stops.

Equalization in DISTANCE is not another word for such rest.

Even when almost no change is visible at one observational scale, internal and external relationships may continue to exist.

Some Momentum may counterbalance one another.

Some may remain effective within a structure.

Some Potential Momentum may remain constrained.

And even a small change in surrounding relationships may alter which Momentum becomes effective.

Therefore:

observable rest $\neq$ completion of Equalization

What we call rest may simply be a state in which effective Momentum does not appear prominently as positional change at a particular observational scale.

Relational possibilities for further change may remain.

Does Equalization Create Order or Disorder?

Another familiar question follows.

Does Equalization make the universe more ordered, or more disordered?

DISTANCE treats the question itself with caution.

“Order” and “disorder” depend partly on what structure the observer chooses as a reference.

When a structure dissolves, we may say that disorder has increased.

But if its components form new relationships and create another structure, a new order may appear elsewhere.

Conversely, a highly organized structure may still maintain strong internal Momentum.

There is therefore no need to define the direction of Equalization simply as either:

order $\rightarrow$ disorder

or

disorder $\rightarrow$ order.

DISTANCE is not primarily concerned with assigning a quantity of order.

Its more direct question is:

Which Momentum are being resolved, and how are relationships being rearranged as a result?

Order and disorder may be observable descriptions of that process at particular scales.

Equalization refers to the more fundamental relational direction of change beneath

them.

The Formation of Structure Can Also Be Part of Equalization

An important consequence follows.

If Equalization is understood merely as dispersion or homogenization, the formation of structure becomes difficult to explain.

Stars form.

Matter aggregates.

Complex structures emerge.

Living systems appear.

At first sight, such processes may seem to be the opposite of smoothing.

Something gathers.

Boundaries form.

Inside and outside become distinguishable.

But once relational Distance is distinguished from geometric distance, the contradiction is no longer necessary.

A process in which components become geometrically closer may itself be a path through which particular Momentum are resolved.

Geometric convergence does not imply decreasing relational Distance.

Instead, multiple relationships may be rearranged into a new configuration through which Momentum can be resolved more effectively.

The formation of structure and Equalization are therefore not necessarily opposing processes.

Structure itself may emerge as a relational rearrangement within the process of Momentum resolution.

This possibility becomes important later when we interpret both the universe and life.

Local Equalization Within a Wider Relational Network

Suppose particular Momentum are being resolved within a structure.

Viewed only from inside that structure, its relationships may appear to be stabilizing.

But the structure itself exists within a wider relational network.

It has external relationships.

There are Differences between it and surrounding structures.

Momentum may exist at broader scales.

Equalization at one level therefore cannot be assumed to be independent of Equalization at larger levels.

The way a local relationship is resolved is influenced by the conditions of wider relationships.

Conversely, local rearrangement changes the conditions of the larger relational network.

Later, this book will gradually develop this distinction using the language of local Momentum and global Momentum.

We do not yet need to develop that architecture in detail.

The important point here is simply that Equalization does not occur only between two isolated states.

Every local Equalization takes place within a wider relational network.

Is Complete Equalization Possible?

Can we imagine a state of complete Equalization in which every Difference has disappeared?

Conceptually, we can.

Every state would be completely equivalent.

No Difference would remain.

No relationship would therefore contain effective Momentum arising from such Difference.

But there is no need yet to assume that this state is the actual destination of the universe.

The universe we observe is instead one in which countless Differences and Momentum are continuously reconfigured.

While one Difference is resolved, the conditions of other Differences change.

When one Momentum weakens, another becomes effective.

Complete Equalization may therefore be better left, at this stage, as a logical limit through which the direction of Equalization can be considered, rather than asserted as a physically attainable final state.

Whether such a limit can actually be reached,

whether it can meaningfully be applied to the universe as a whole,

or whether the complexity of the relational network makes it permanently unreachable, need not be decided here.

Equalization Is a Continuing Process, Not an Endpoint

We can now state the meaning of Equalization more clearly.

Equalization is not simple homogenization that makes everything one state.

It is not a process that eliminates every Difference at once.

It is not the minimization of Distance.

And it is not a process whose goal is to stop all motion.

Equalization is:

the continuing process in which Momentum generated through effective Difference is resolved, and that resolution reconfigures the relational network, thereby changing which Momentum becomes effective next.

Equalization therefore contains two inseparable aspects.

One is resolution.

Existing Momentum is resolved.

The other is reconfiguration.

That resolution changes the relational network and makes other Momentum effective.

Resolution produces reconfiguration.

Reconfiguration establishes new conditions for resolution.

For this reason, Equalization is not a single event.

It is the continuing direction of change.

From Difference to Equalization

We can now place the concepts introduced in Chapter 1 into one sequence.

First, there is Difference.

States are not equivalent.

That Difference has some degree of effectiveness within an actual relationship.

We express the effective nearness or farness of that relationship as Distance.

A strongly effective relationship exhibits directionality of change.

That directionality is Momentum.

Momentum is then resolved within the relational network.

We call the continuing process of resolution and reconfiguration Equalization.

The basic conceptual flow of Chapter 1 can therefore be written as:

Difference $\rightarrow$ Distance $\rightarrow$ Momentum $\rightarrow$ Equalization $\rightarrow$ Momentum resolution and increasing Distance $\rightarrow$ relational rearrangement $\rightarrow$ newly effective Momentum

There is no fixed final point in this sequence.

Relational rearrangement changes the conditions under which new Difference, Distance, and Momentum become effective.

The process is therefore closer to a repeatedly reconfigured relational cycle than to a simple straight line.

Yet it is not a cycle that returns to exactly the same state.

Every resolution changes the relational network.

Looking Again at Time and Space

We can now return to the beginning of Chapter 1.

We did not begin this discussion in order to deny time and space.

Rather, because these concepts are so familiar that we routinely take them for granted as the starting framework of explanation, we temporarily placed them in the background.

What remained was change and relationship.

Looking more closely at change revealed Difference.

Considering the effectiveness of Difference within relationships required Distance.

The directionality through which an effective relationship appears in actual change was called Momentum.

And the continuing process through which Momentum is resolved and reconfigured was called Equalization.

Seen from this perspective, it becomes possible to describe a basic logic of change without first placing time and space as independent stages upon which everything else occurs.

This does not require us to conclude that time does not exist.

Nor must we declare space an illusion.

A more careful question is enough:

How are what we call time and space related to the way these relational changes are observed?

Perhaps time is not an empty container into which change is placed, but a scale through which the intensity of change is read.

Perhaps space is not simply an empty stage upon which relationships are placed, but a geometric way of reading relational complexity.

There is no need to settle those questions prematurely here.

The purpose of Chapter 1 has not been to complete a new theory of spacetime.

It has been to ask whether, once our familiar starting assumptions are temporarily set aside, the logic of change can still be reconstructed through the relational language of Difference, Distance, Momentum, and Equalization.

Now We Look Again at the Universe

These concepts will have little significance if they remain only an abstract vocabulary.

They must also work consistently when applied to the actual universe.

How should the early universe be interpreted?

Must what we call cosmic expansion be understood only as the expansion of space itself?

If countless relationships changed within an extremely dense initial state, can that process be reread in terms of Momentum resolution and changing Distance?

Can the separation of matter, the formation of structures, and the emergence of the universe we observe today be interpreted as continuing relational rearrangement through Equalization?

These are no longer merely questions of terminology.

The concepts of DISTANCE must now be applied to the history of the universe itself.

In Chapter 1, we prepared the language of explanation.

Beginning with the next chapter, we use that language to look at the universe again.

If Difference exists, if Distance expresses relational effectiveness, if Momentum reveals the directionality of that relationship, and if Equalization resolves Momentum while continually reconfiguring relationships—

then where, within that continuing process, is the universe we observe today?

That is where the next chapter begins.